

Statistical Modeling With Label Constraint for Radar Target Recognition

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Motivated by the problem of radar target recognition, we develop a label-aided factor analysis (LA-FA) model for statistical modeling of high-resolution range profile (HRRP) under the prerequisite that the HRRP data are Gaussian distributed. The LA-FA model is the extension of the multitask learning-based factor analysis (MTL-FA) model, which is mainly applied to the recognition problem with small training data size. Compared to the MTL-FA model, our LA-FA model introduces the discrete class labels via Sigmoid-Bernoulli hierarchy to restrict the learning of model parameters, which offers the potential to enhance the separability of statistical models from different classes, thus beneficial to the improvement of recognition capability. In addition, since the noise level of a test sample is usually different from those of the training samples in the real application, we introduce a noise-robust modification method for Gaussian-based models. The proposed modification method is implemented by updating the noise level parameter of the statistical model according to the estimated signal-to-noise ratio (SNR) of test HRRP. Experiments on measured HRRP data demonstrate the better recognition performance of the LA-FA model with limited training data and our noise-robust model modification method under low test SNR. Especially, when there are 20 training HRRP samples per frame, the recognition rate of our LA-FA model is 7% higher than that of the MTL-FA model, and moreover, the recognition accuracy of the noise-robust LA-FA model is 3% higher than that of the LA-FA model without modification under the condition of SNR = 15 dB.

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I. INTRODUCTION

With the application of information and intelligence technologies in modern warfare, target recognition has become a research hotspot in radar area. Wideband radar signals can provide abundant target structure information, and thus can be used commonly as a characteristic feature for radar target recognition. Due to the property of easy acquisition and processing, radar high-resolution range profile (HRRP) has been extensively studied and successfully applied to radar target recognition community [3]–[9], [12], [17], [19]–[21].

The electromagnetism characteristic of the target can be described by the scattering center model [14]. In addition, since the target size is much larger than the range resolution cell for the high-resolution radar system, a target does not appear as a “point target” any longer, but consists of many scatterers distributed in several range cells along the radar line-of-sight. An HRRP is the amplitude of the coherent summations of complex returns from target scatterers in each range cell, as shown in Fig. 1.

The existing methods for HRRP-based radar target recognition can be classified into two main categories: nonparametric method and parametric method. In the nonparametric method, statistical features or transform domain features of HRRP data are extracted without the parametric modeling, and then an appropriate classifier is chosen to classify the features. Since the performance of the recognition system is heavily dependent on the extracted features, the research of nonparametric radar HRRP target recognition mostly focuses on the feature extraction. In [16] and [17], some physically meaningful features, such as target size, center of mass, and the number of the peaks, are extracted from the HRRP data. The central moment is calculated and applied to radar target recognition in [18]. Some researchers [19] investigate the bispectra feature and Du *et al.* [5], [29] analyze the recognition performance of power spectrum, fast Fourier transform-magnitude feature, and various high-order spectra features. Nevertheless, the feature extraction in above methods is separate from the training of the classifier, which often leads to the mismatch between the features and the back-end classifier, thus limiting the recognition performance. In the parametric method, a parametric model is built for HRRP data. Based on the constructed model, the latent features of observations can be revealed and used for recognition task. The PCA-based feature subspace is constructed for radar HRRP target recognition in [20]. Feng *et al.* [21] utilize dictionary learning to represent the favor sparse overcomplete features of HRRP data. In [7], the stable hierarchical features are learned via the stacked corrective autoencoders, which adds the average profile as the correction term to the autoencoder framework. Compared to the nonparametric feature extraction, these methods are data-driven and less depend on the researchers' experience. Furthermore, they are capable of extracting more abstract features hidden in data. However, they still suffer the problem of the mismatch between the features and the back-end classifier, due to the

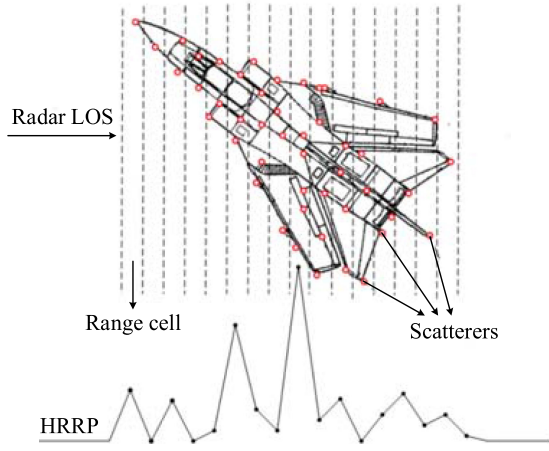


Fig. 1. Illustration of an HRRP sample from a plane target, where the circles on the plane represent the scatterers. This figure is cited from [12].

independence of feature extraction and classifier training. Besides the latent feature mining, the parametric model can also be applied to radar HRRP statistical recognition. In HRRP-based statistical recognition, after determining the form of data distribution, a parametric model is constructed to estimate the distribution parameters. According to the estimated parameters, a Bayes classifier is used to recognize the test HRRP sample. By Bayes classifier we mean that the class membership of a measurement HRRP sample $\mathbf{x}$ is determined based on the posterior probabilities of all class memberships $p(c|\mathbf{x})$ ($c \in \{1, 2, \dots, C\}$ denotes the class membership). Statistical recognition has the advantages of giving a measure of confidence for the class predictions, and the opportunity to combine the probabilities with additional information [2], [22]. Furthermore, the Bayes classifier is constructed via the learned model parameters, which avoids the mismatch of the feature and classifier. In this paper, we mainly study the parametric modeling for radar HRRP statistical recognition.

As discussed above, we first need to assume the distribution of the HRRP sample in radar HRRP statistical recognition. According to the fluctuating target model by Swerling [24], each range cell of an HRRP is Rayleigh distributed or Ricean distributed under the ideal condition. The ideal condition requires a large number of the scatterers with almost the same intensities per range cell, which ensures that the central limit theorem holds. Unfortunately, increasing the resolution implies a reduction of the number of scatterers per range cell; thus, the Rayleigh distribution and Ricean distribution tend to fail for the echo of a range cell under the high-resolution condition. As a compromise between accuracy and computation resource, Gaussian distribution has been widely adopted in radar HRRP statistical recognition [4]–[6], [25]–[27]. Fig. 2 shows the approximated probability density functions (PDFs) of echoes in some typical range cells of three targets, from which we can find that the probability distribution approximated by histogram method is close to that estimated by Gaussian distribution.

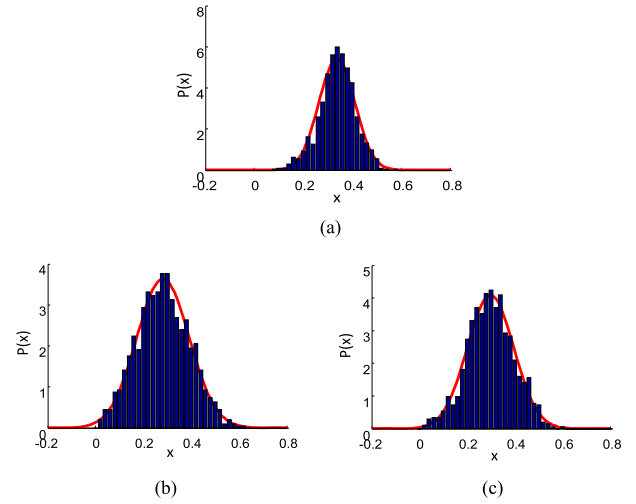


Fig. 2. PDFs of echoes in some typical range cells of three targets. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26. The blue bars represent the PDF of the echoes from the range cell approximated by the histogram method, and the red line represents the PDF estimated by the Gaussian distribution.

According to the Bayes theorem, we have $p(c|\mathbf{x}) \propto p(\mathbf{x}|c)p(c)$, where $p(\mathbf{x}|c)$ is the class-conditional likelihood and $p(c)$ is the prior class probability. In general, there are two different approaches to assign priors: subjective and objective approaches [10]. The subjective approach is to encapsulate as much prior knowledge as possible into the prior probability. In this approach, the prior knowledge may be due to previous experiments or expert knowledge and is often reliable. The objective approach is used when the modeler has no available prior knowledge. In this case, the prior probability should be noninformative, that is, the difference of the class posteriors is determined by that of the class-conditional likelihoods alone. In our work, we have no prior knowledge about the class membership for an HRRP sample $\mathbf{x}$. Thus, the noninformative prior is imposed on $p(c)$, i.e., $\mathbf{x}$ has the equal prior probability of occurrence in each target. Furthermore, the estimation of the posterior probability is turned into that of the class-conditional likelihood.

Based on the scattering center model, the scatterer distribution of a target will vary with the target aspect, i.e., the angle between the directions of target's centerline and the radar LOS, thus causing the sensitivity of HRRP to the change of target-aspect. Furthermore, when there is a considerable target-aspect variation, some scatterers of the target will remove from one range cell to another, i.e., scatterers' motion through range cells (MTRC). In Fig. 3(a1) and (b1), we present two sets of HRRP samples from Yark-42, where only 161 range cells close to the support for each HRRP are shown for clarity, and the corresponding target-aspects of the two HRRP sets change from 2.16° to 3.01° and from 4.08° to 4.93° , respectively. Since the critical target-aspect angle for avoiding scatterers' MTRC is about 0.86° for Yark-42, and thus, the scatterers' MTRC does not occur in each HRRP set (the

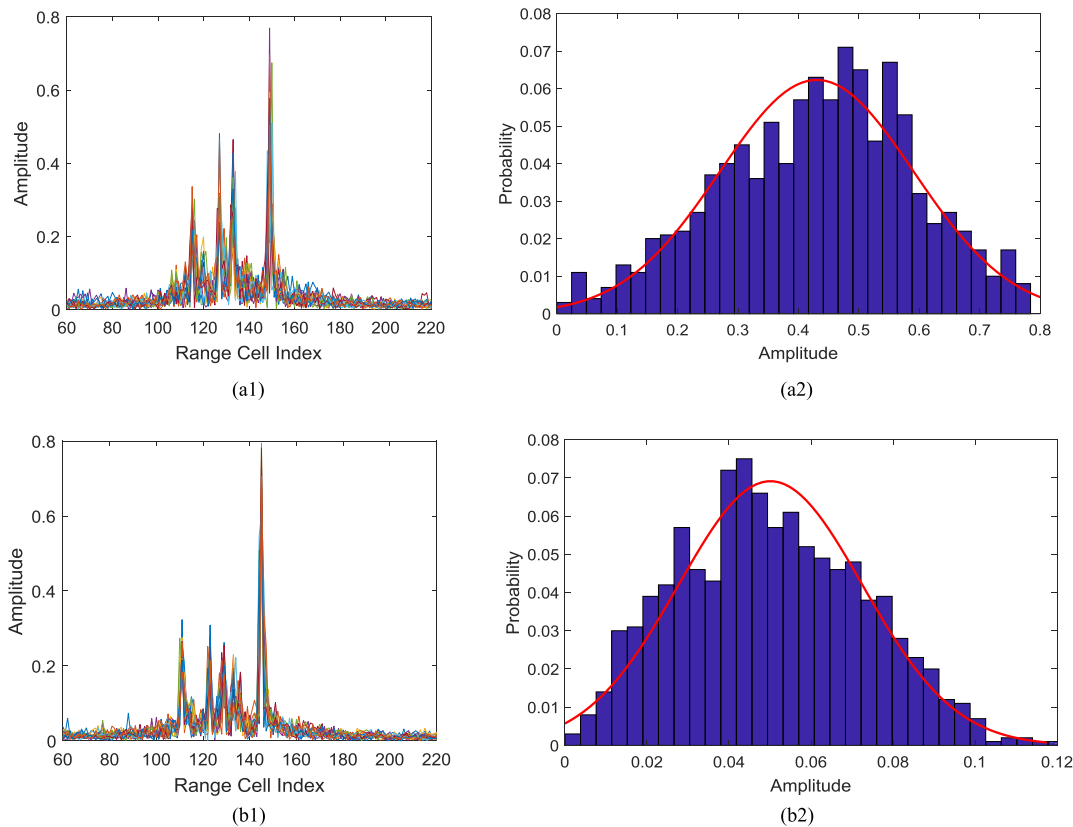


Fig. 3. Two sets of HRRP samples from Yark-42 and their corresponding PDFs of echoes from the 149th range cell. (a1) Set of HRRP samples with the change interval of target-aspect being $2.16^\circ - 3.01^\circ$. (b1) PDF of echoes from the 149th range cell of HRRPs in (a1). (a2) Set of HRRPs with the change interval of target-aspect being $4.08^\circ - 4.93^\circ$. (b2) PDF of echoes from the 149th range cell of HRRPs in (a2).

corresponding target-aspect change of each set is 0.85°), but occurs between the two HRRP sets. In Fig. 3(a2) and (b2), we show the PDFs of echoes from the 149th range cell of two HRRP sets. In Fig. 3(a1) or (b1), when there is nonoccurrence of scatterers' MTRC, we see that the locations and fluctuation characteristics of HRRP peaks change small. By comparing Fig. 3(a1) and (b1), when the scatterers' MTRC occurs, we see that the peak locations and fluctuation characteristics change a lot, which leads to the great change of statistical distribution of HRRPs, as shown in Fig. 3(a2) and (b2). If we use a model to describe all samples from a target without considering scatterers' MTRC, the recognition performance may be weakened, due to much different statistical characteristics among HRRPs with greater target-aspect variations. To overcome the above problem, the HRRP dataset from a target is divided into several subsets, where HRRPs in each subset are collected from a target-aspect sector roughly without scatterers' MTRC, and each subset is defined to be an aspect-frame. Here "roughly" means that the critical angle of the target-aspect change corresponding to the nonoccurrence of scatterers' MTRC is approximately calculated in the experiment, and it is not an exact value. After aspect-frame partition, we learn a statistical model for each frame of each target. Therefore, for radar HRRP statistical recognition, we first estimate the frame-conditional likelihood. Accord-

ing to $p(\mathbf{x}|c) = \max_m p(\mathbf{x}|m, c)$ ($m \in \{1, \dots, M_c\}$ denotes the frame membership, and M_c denotes the number of the aspect-frames in the c th target), the class-conditional likelihood can then be obtained.

Some Gaussian-based parametric models, such as principal component analysis-based subspace approximation model [4], probabilistic principal component analysis (PPCA) model [4], Gaussian model [8], and factor analysis (FA) model [4], [13] are developed to describe the frame-conditional likelihood of HRRP data. All these statistical models are constructed for each aspect-frame of each target, i.e., single-task learning (STL), and of high complexity. Therefore, when there are sufficient training data in each frame, they can accurately describe the statistical characteristics of HRRP data and achieve good recognition performance.

However, in the real application, due to the limited radar sampling rate (the number of HRRP samples collected by radar per unit time) and great change of flight attitude for the target with high velocity, it is hard to collect enough HRRP samples in each aspect-frame. The decrease of training data often leads to the reduction of parameter estimation precision. When the number of training samples decreases to a certain extent, there is a large estimation error, and in such a situation, we think that the training data are of small size. In the case of small training data size, the "small

size problem” [23] occurs in the above STL-based model, such as weak recognition ability, poor generalization performance, and so on. To address this problem, our recent work [5] applied the thought of multitask learning (MTL) [1] to radar HRRP statistical recognition and developed an MTL factor analysis (MTL-FA) model, wherein all samples are mapped to their corresponding latent factor variables via a shared factor dictionary. By information sharing, the MTL-FA model has a fewer parameters to be estimated and achieves effective recognition performance with limited training data. Nevertheless, the MTL-FA model only focuses on the description of observations. When there are only a few observed samples, the information contained in the observations is limited. Therefore, although the MTL-FA model can accurately describe the observed data, its performance is still limited. Considering this limitation, we introduce the constraint information into MTL-FA model and propose a label-aided factor analysis (LA-FA) model to further improve the recognition accuracy with small training data size.

In our LA-FA model, the latent factor variables from different classes are also mapped to their corresponding class labels via Sigmoid-Bernoulli hierarchy. By introducing the label constraint, the difference of latent factor variables from different classes increases, which provides the potential to enhance the separability of statistical models among different classes. Thus, the recognition performance can be improved.

Statistical models are typically constructed with the assumption that the training data and test data are obtained under similar measurement circumstances. In actual application, training HRRP data are collected under the condition of high signal-to-noise ratio (SNR) via some cooperative measurement experiments, whereas test samples are usually obtained in the noncooperative circumstance, where the high SNR cannot be guaranteed. Since the noise component can change the statistical distribution of the noisy test HRRP sample, the statistical characteristic of the test HRRP sample will remarkably differ from that of the model trained under high SNR when the test SNR decreases to a certain extent, thus causing the serious degradation of recognition performance. In this case, we think that the test HRRP sample is obtained under low SNR condition. Considering the weak recognition ability of the statistical model in the case of low test SNR, we introduce a noise-robust modification method for Gaussian-based models. The proposed modification method can update the noise level parameter of the model according to the estimated SNR of the test sample in the classification stage. As a result, the statistical model with modification can better match the noisy test sample and have the ability to obtain higher recognition accuracies under test condition of low SNR.

We summarize the main contributions of our work as follows:

- 1) We incorporate label information into the FA model based on MTL mechanism and propose a novel Gaussian statistical model, i.e., LA-FA, where the class labels

TABLE I
List of Notations in the LA-FA MODEL

Notations	Meanings of notations	Notations	Meanings of notations
C	The number of classes (targets)	$\mathbf{w}_n^{c,m} \in \mathbb{R}^K$	Latent factor variable representing the weight of $\mathbf{x}_n^{c,m}$
M_c	The number of frames for target c	$\mathbf{z}_n^{c,m} \in \mathbb{R}^K$	Latent factor variable selecting the columns of $\mathbf{A}$
N	Training sample number per frame	$\Phi_n^{c,m} \in \mathbb{R}^L$	Label reserving constraint term
P	Dimension of observed sample	$\boldsymbol{\epsilon}_n^{c,m} \in \mathbb{R}^P$	Noise variable corresponding to $\mathbf{x}_n^{c,m}$
L	Dimension of class label	$\boldsymbol{\eta}_n^{c,m} \in \mathbb{R}^L$	Noise variable corresponding to $\mathbf{b}_n^{c,m}$
K	Dimension of latent factor variable	$\alpha^{c,m}$	Covariance precision of $\mathbf{w}_n^{c,m}$
$\mathbf{x}_n^{c,m} \in \mathbb{R}^P$	The n th observed sample from m th frame of target c	$\mathbf{u}^{c,m}$	Prior that controls the sparsity of $\mathbf{z}_n^{c,m}$
$\mathbf{b}_n^{c,m} \in \mathbb{R}^L$	Class label of $\mathbf{x}_n^{c,m}$	$\gamma^{c,m}$	Covariance precision of $\boldsymbol{\epsilon}_n^{c,m}$
$\boldsymbol{\mu}^{c,m} \in \mathbb{R}^P$	Mean vector of the m th frame of target c	$\lambda^{c,m}$	Covariance precision of $\boldsymbol{\eta}_n^{c,m}$
$\mathbf{A} \in \mathbb{R}^{P \times K}$	Shared factor dictionary	$a_0 \dots h_0$	Hyper-parameters of model
$\mathbf{D} \in \mathbb{R}^{L \times K}$	Label preserving dictionary	$\delta_n^{c,m}$	Augmented variable

of observed data are introduced via Sigmoid-Bernoulli hierarchy. With label constraint, our LA-FA model can achieve better recognition performance under limited training data.

- 2) Data augmentation is exploited to make the derivation of model parameters tractable, when we implement our LA-FA model via Gibbs sampling.
- 3) A noise-robust modification method is introduced for Gaussian-based statistical models to improve the recognition performance under the test condition of low SNR.

In Table I, we list all notations in our LA-FA model to help readers to understand the dimensions of matrices and definitions of notations clearly.

The remainder of our paper is organized as follows. We introduce the LA-FA model for HRRP data in Section II. The radar target recognition based on LA-FA model is discussed in Section III. The detailed experimental results based on measured HRRP dataset are provided in Section IV. The conclusions are given in Section V.

II. LA-FA MODEL FOR HRRP DATA

A. Review of FA and MTL-FA Models

As discussed in Introduction, the aspect-frame partition should be implemented according to the nonoccurrence of scatterers' MTRC before modeling the HRRP data. The critical angle of target-aspect change for avoiding

scatterers' MTRC is

$$(\Delta\varphi)_{\text{MTRC}} \stackrel{\text{Def}}{=} \frac{c}{2BR} \quad (1)$$

where B , R , and c are the bandwidth of radar signal after range compression, the target dimension in cross range and the speed of light, respectively. Based on (1), we divide the HRRP dataset from a target into several frames, where the corresponding target-aspect change of HRRPs in each frame is $(\Delta\varphi)_{\text{MTRC}}$, in the experiment. For example, for Yark-42, the bandwidth of radar signal B is 400 MHz, the speed of light c is 3×10^8 m/s, and the target dimension in cross range R is about 25 m. Thus, $(\Delta\varphi)_{\text{MTRC}} \approx 0.86^\circ$. In addition, the corresponding target-aspect of each HRRP can be obtained according to the radar tracking information. Based on the above analysis, we can divide the whole HRRP dataset from Yark-42 into 35 frames, and the target-aspect change of each frame is 0.86° .

After aspect-frame partition, Du *et al.* [4] construct an FA model for each frame of each target to describe the statistical characteristic of the frame HRRP data. FA model is a probability generative model to depict the observation with high dimension distributed in low latent subspace. Let $\mathbf{X}^{c,m} = \{\mathbf{x}_n^{c,m}\}_{n=1}^N$ ($\mathbf{x}_n^{c,m} \in \mathbb{R}^P$) represent the HRRPs from the m th frame of the c th target, where $c = 1, 2, \dots, C$ denotes class membership with C denoting the number of targets, $m = 1, 2, \dots, M_c$ denotes frame membership with M_c denoting the number of the frames in target c , and N denotes the number of HRRP samples in each frame. The generative FA model can be represented as follows:

$$\mathbf{x}_n^{c,m} = \mathbf{A}^{c,m} \mathbf{w}_n^{c,m} + \boldsymbol{\mu}^{c,m} + \boldsymbol{\varepsilon}_n^{c,m} \quad (2)$$

where $\boldsymbol{\mu}^{c,m} \in \mathbb{R}^P$ is the mean vector and can be estimated by $\boldsymbol{\mu}^{c,m} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n^{c,m}$; $\mathbf{A}^{c,m} = [\mathbf{a}_1^{c,m}, \dots, \mathbf{a}_K^{c,m}] \in \mathbb{R}^{P \times K}$ is the factor dictionary with K denoting the number of factors ($P \gg K$) and each column $\mathbf{a}_k$ ($k = 1, \dots, K$) being called a dictionary atom; $\mathbf{w}_n^{c,m} \in \mathbb{R}^K \sim N(0, \mathbf{I}_K)$ is the latent factor variable with $N(\cdot)$ denoting the Gaussian distribution and $\mathbf{I}_K$ being the $K \times K$ dimensional identity matrix; $\boldsymbol{\varepsilon}_n^{c,m} \in \mathbb{R}^P \sim N(\mathbf{0}, \Psi^{c,m})$ is the noise variable with $\Psi^{c,m}$ being a diagonal matrix and the diagonal elements being different. Then, we can obtain the estimation of PDF from the m th frame of target c as follows:

$$p(\mathbf{x}_n^{c,m}) = N(\mathbf{x}_n^{c,m} | \boldsymbol{\mu}^{c,m}, \mathbf{A}^{c,m} \mathbf{A}^{c,mT} + \Psi^{c,m}). \quad (3)$$

From (2), we know that the production of factor dictionary $\mathbf{A}^{c,m}$ and latent factor variable $\mathbf{w}_n^{c,m}$ reflects the deviation of observed data $\mathbf{x}_n^{c,m}$ from $\boldsymbol{\mu}^{c,m}$. In addition, each dictionary atom $\mathbf{a}_k^{c,m}$ represents a basis and a sample $\mathbf{x}_n^{c,m}$ is the linear combination of the bases with weight vector, i.e., latent factor variable $\mathbf{w}_n^{c,m}$. Based on the above analysis, the dictionary atoms of the factor dictionary embody the dispersion of observations relative to the mean vector, and thus, they capture the information reflecting the frame covariance matrix, as shown in (3). In Fig. 4, we present the difference vectors between HRRP samples and the corresponding mean vector from three aspect-frames of Yark-42. For each subfigure of Fig. 4, different colored

lines represent difference vectors corresponding to different HRRPs, and all the difference vectors are superimposed which can reflect the dispersion of HRRP samples relative to the frame mean vector. As shown in Fig. 4, there are significant discrepancies in the dispersion of HRRPs relative to the mean vector for different frames, which implies that the covariance matrices from different frames should be of great difference. Accordingly, the learned factor dictionary of each frame should be distinguishing and so is the latent factor variable. That is, the learned model parameters (factor dictionary and latent factor variables) of each frame can capture the information associated with the frame covariance matrix in the FA model.

In the training phase in [4], an FA model is learned for each frame of each target, i.e., STL-FA model. The parameter estimations, including the posterior estimation of latent factor variable $\mathbf{w}_n^{c,m}$ and the point estimations of factor dictionary $\mathbf{A}^{c,m}$ and noise covariance matrix $\Psi^{c,m}$, can be implemented via the iterative expectation-maximization algorithm [10], [11].

We usually use the training data as much as possible to obtain the reliable estimators of unknown parameters in a statistical model. However, in the real application, substantial training data cannot be easily obtained. Actually, the HRRPs from different targets have certain similarity. Therefore, for all frames of all targets, their corresponding dictionary atoms to generate the observation space should have certain overlaps. Motivated by this, Du *et al.* [5] develop an MTL-FA model. The MTL-FA model uses a large-sized dictionary to describe all the HRRPs, which offers the opportunity to share some dictionary atoms among all frames of all targets. As a result, the number of dictionary atoms to be estimated greatly decreases and the recognition performance can be improved under the condition of limited training samples per frame. In detail, the MTL-FA model for the m th frame of the c th target is represented as follows:

$$\mathbf{x}_n^{c,m} = \mathbf{A} (\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}) + \boldsymbol{\mu}^{c,m} + \boldsymbol{\varepsilon}_n^{c,m} \quad (4)$$

where $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_K] \in \mathbb{R}^{P \times K}$ is the shared factor dictionary for all frames of all targets. For each frame, only a small amount of dictionary atoms in $\mathbf{A}$ are used to represent the frame HRRP samples, and the subset of dictionary atoms in each frame is different. To automatically select the dictionary atoms for each frame, a binary vector $\mathbf{z}_n^{c,m} = [z_{n,1}^{c,m}, \dots, z_{n,K}^{c,m}]^T$ with $z_{n,k}^{c,m} \in \{0, 1\}$ is introduced in (4). If there are nonzero elements in $\{z_{n,k}^{c,m}\}_{k=1}^K$, the k th dictionary atom $\mathbf{a}_k$ is selected for representing the frame m of target c ; otherwise, $\mathbf{a}_k$ is not utilized. Then, $\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m} \in \mathbb{R}^K$ can be considered as the latent factor variable of $\mathbf{x}_n^{c,m}$ with $\odot$ representing the Hadamard vector product.

Based on the above analysis, $z_{n,k}^{c,m}$ is a binary variable. Thus, we should impose a Bernoulli prior on $z_{n,k}^{c,m}$ to depict the distribution characteristic of $z_{n,k}^{c,m}$. Similar to the STL-FA model, the MTL-FA model is implemented within a Bayesian framework, and the prior distributions for all the

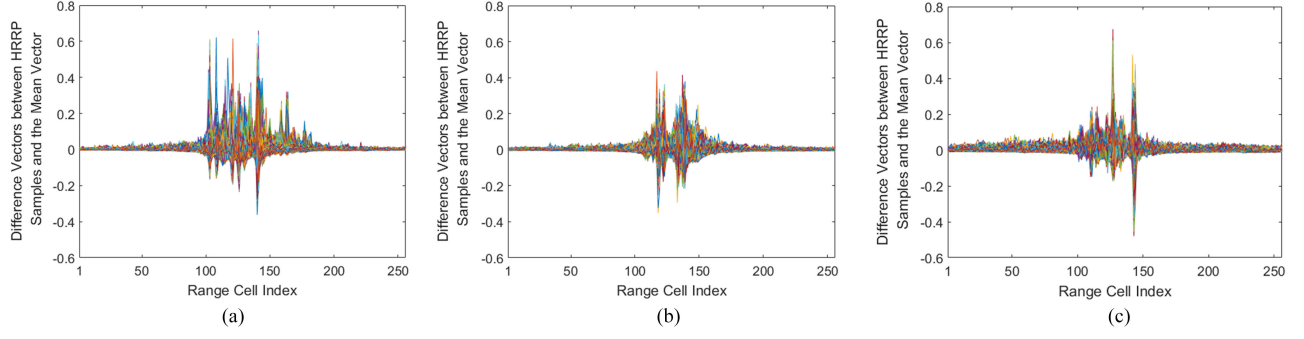


Fig. 4. Difference vectors between HRRPs and the corresponding mean vectors from three different frames of Yark-42, with different colored lines represent difference vectors corresponding to different HRRPs in each subfigure. (a) First frame. (b) Fourteenth frame. (c) Thirty-fourth frame.

parameters are as follows:

$$\begin{aligned} \mathbf{a}_k &\sim N(\mathbf{0}, 1/P\mathbf{I}_P), \mathbf{w}_n^{c,m} \sim N(\mathbf{0}, 1/\alpha^{c,m}\mathbf{I}_K), \\ \alpha^{c,m} &\sim \text{Ga}(a_0, b_0), z_{n,k}^{c,m} \sim \text{Bern}(u_k^{c,m}), \\ u_k^{c,m} &\sim \text{Beta}(c_0, d_0), \epsilon_n^{c,m} \sim N(\mathbf{0}, 1/\gamma^{c,m}\mathbf{I}_P), \\ \gamma^{c,m} &\sim \text{Ga}(e_0, f_0), \end{aligned} \quad (5)$$

where $a_0, b_0, c_0, d_0, e_0, f_0$ are preset hyperparameters, and $\text{Ga}(\cdot)$, $\text{Bern}(\cdot)$, $\text{Beta}(\cdot)$ are referred to as Gamma, Bernoulli, and Beta distributions, respectively.

Compared to the STL-FA model, the MTL-FA model imposes a Gaussian prior distribution on the factor dictionary and also imposes Gamma priors on the distributions of latent factor variable and noise variable. It is motivated by the fact that more accurate parameter estimation can be obtained by introducing more and reasonable prior knowledge in the case of small data learning. In addition, since the shared dictionary $\mathbf{A}$ is of large size and only a few dictionary atoms are selected for each frame, $\mathbf{z}_n^{c,m}$ should be sparse. Therefore, a Beta distribution is imposed on the distribution of $z_{n,k}^{c,m}$. By this Beta process structure, we can control the sparsity of $\mathbf{z}_n^{c,m}$.

B. LA-FA Model

The work in [5] shows that the MTL-FA model has the better recognition performance than the STL-FA model under the condition of limited training samples, due to the more accurate parameter estimation of the MTL-FA model. Nevertheless, the MTL-FA model only focuses on the description of observations. When there are only a few observed samples, the information contained in observations is limited. Therefore, although the MTL-FA model can accurately describe observed data, its performance is still limited. To further improve the model recognition rate, we incorporate additional information, i.e., class label of the sample, into the MTL-FA and propose a LA-FA model, which takes advantage of the differences among class labels to enhance the separability of statistical models from different classes.

We define that the class labels of all HRRP samples from the first target are $[1, 0, \dots, 0]^T \in \mathbb{R}^L$, and all the class labels from the second target are $[0, 1, \dots, 0]^T \in \mathbb{R}^L$. The

class labels of the remainder targets are defined by analogy. Then, our LA-FA model can be expressed as follows:

$$\begin{aligned} \mathbf{x}_n^{c,m} &= \mathbf{A}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}) + \mu^{c,m} + \epsilon_n^{c,m} \\ \mathbf{b}_n^{c,m} &\sim \text{Bern}(\sigma(\boldsymbol{\varphi}_n^{c,m})) \\ \boldsymbol{\varphi}_n^{c,m} &= \mathbf{D}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}) + \boldsymbol{\eta}_n^{c,m} \end{aligned} \quad (6)$$

where the expression of the first line is the same as that of the MTL-FA model in (4). $\mathbf{b}_n^{c,m} \in \mathbb{R}^L$ denotes the class label of $\mathbf{x}_n^{c,m}$ and obeys Bernoulli distribution with Bernoulli parameter $\sigma(\boldsymbol{\varphi}_n^{c,m}) = \exp(\boldsymbol{\varphi}_n^{c,m}) / (1 + \exp(\boldsymbol{\varphi}_n^{c,m}))$ and $\sigma(\cdot)$ denoting Sigmoid transformation. According to the property of Bernoulli distribution and the monotonically increasing property of the sigmoid function, the element values in $\boldsymbol{\varphi}_n^{c,m}$ that correspond to the nonzero elements of $\mathbf{b}_n^{c,m}$ are much larger than those in $\boldsymbol{\varphi}_n^{c,m}$ that correspond to the zero elements of $\mathbf{b}_n^{c,m}$. That is, a one-to-one correspondence exists between the element values of $\mathbf{b}_n^{c,m}$ and $\boldsymbol{\varphi}_n^{c,m}$, i.e., the differences of class labels are preserved in $\{\boldsymbol{\varphi}_n^{c,m}\}_{n=1, m=1, c=1}^{N, M_c, C}$, and furthermore, $\boldsymbol{\varphi}_n^{c,m}$ can be used to restrict the parameters' learning to strengthen the difference of models from different classes. Thus, we refer to $\boldsymbol{\varphi}_n^{c,m}$ as the label preserving constraint term. $\mathbf{D} \in \mathbb{R}^{L \times K}$ denotes the label preserving dictionary that maps the latent factor variable $\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}$ to label $\mathbf{b}_n^{c,m}$.

Similar to the settings of priors in the MTL-FA model, we also impose some special prior distributions on the parameters or the distributions of parameters in our LA-FA model for an easier implementation. All the parameters' priors in our LA-FA model are set as follows:

$$\begin{aligned} \mathbf{w}_n^{c,m} &\sim N(\mathbf{0}, 1/\alpha^{c,m}\mathbf{I}_K), \alpha^{c,m} \sim \text{Ga}(a_0, b_0), \\ \mathbf{z}_{n,k}^{c,m} &\sim \text{Bern}(u_k^{c,m}), u_k^{c,m} \sim \text{Beta}(c_0, d_0), \\ \mathbf{a}_k &\sim N(\mathbf{0}, 1/P\mathbf{I}_P), \epsilon_n^{c,m} \sim N(\mathbf{0}, 1/\gamma^{c,m}\mathbf{I}_P), \\ \gamma^{c,m} &\sim \text{Ga}(e_0, f_0), \mathbf{d}_k \sim N(\mathbf{0}, 1/L\mathbf{I}_L), \\ \boldsymbol{\eta}_n^{c,m} &\sim N(\mathbf{0}, 1/\lambda^{c,m}\mathbf{I}_L), \lambda^{c,m} \sim \text{Ga}(g_0, h_0) \\ \boldsymbol{\varphi}_n^{c,m} &\sim N(\mathbf{D}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}), 1/\lambda^{c,m}\mathbf{I}_L). \end{aligned} \quad (7)$$

A graphical representation of the LA-FA model is depicted in Fig. 5.

From (6) and (7), we see that the latent factor variable $\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}$ is the “bridge” of the observed sample $\mathbf{x}_n^{c,m}$

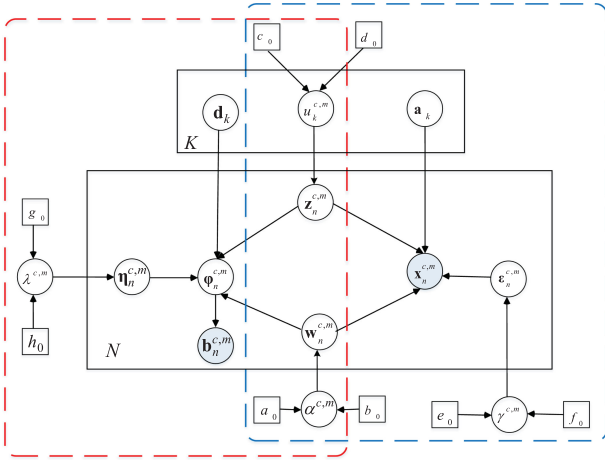


Fig. 5. Graph model of the LA-FA model. The left section shows the label-aided module, and the right shows the factor analysis module.

and class label $\mathbf{b}_n^{c,m}$. Thus, the estimations of $\mathbf{w}_n^{c,m}$ and $\mathbf{z}_n^{c,m}$ will be affected by both $\mathbf{x}_n^{c,m}$ and $\mathbf{b}_n^{c,m}$. In detail, $\mathbf{w}_n^{c,m}$ is sampled from the posterior distribution $p(\mathbf{w}_n^{c,m} | \mathbf{x}_n^{c,m}, \varphi_n^{c,m})$. According to the Bayes theorem, $p(\mathbf{w}_n^{c,m} | \mathbf{x}_n^{c,m}, \varphi_n^{c,m})$ can be expressed as follows:

$$\begin{aligned} p(\mathbf{w}_n^{c,m} | \mathbf{x}_n^{c,m}, \varphi_n^{c,m}) &\propto p(\mathbf{x}_n^{c,m} | \mathbf{w}_n^{c,m}) \\ &p(\varphi_n^{c,m} | \mathbf{w}_n^{c,m}) p(\mathbf{w}_n^{c,m}). \end{aligned} \quad (8)$$

For $p(\mathbf{x}_n^{c,m} | \mathbf{w}_n^{c,m})$ given other model parameters including $\mathbf{A}$, $\mathbf{z}_n^{c,m}$, $\mu^{c,m}$, and the prior distribution $\varepsilon_n^{c,m} \sim N(0, 1/\gamma^{c,m} \mathbf{I}_P)$, we have

$$\begin{aligned} p(\mathbf{x}_n^{c,m} | \mathbf{w}_n^{c,m}) &= (2\pi)^{-\frac{P}{2}} |1/\gamma^{c,m} \mathbf{I}_P|^{-\frac{1}{2}} \\ &\times \exp\left(-\frac{\gamma^{c,m}}{2} \mathbf{x}_n^{c,mT} \mathbf{x}_n^{c,m}\right) \end{aligned} \quad (9)$$

where $\mathbf{x}_n^{c,m} = \mathbf{x}_n^{c,m} - \mathbf{A}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}) - \mu^{c,m}$. Similarly, for $p(\varphi_n^{c,m} | \mathbf{w}_n^{c,m})$

$$\begin{aligned} p(\varphi_n^{c,m} | \mathbf{w}_n^{c,m}) &= (2\pi)^{-\frac{L}{2}} |1/\lambda^{c,m} \mathbf{I}_L|^{-\frac{1}{2}} \\ &\times \exp\left(-\frac{\lambda^{c,m}}{2} \varphi_n^{c,mT} \varphi_n^{c,m}\right) \end{aligned} \quad (10)$$

where $\varphi_n^{c,m} = \varphi_n^{c,m} - \mathbf{D}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m})$. The label preserving constraint term $\varphi_n^{c,m}$ can be sampled from its posterior distribution $p(\varphi_n^{c,m} | \mathbf{b}_n^{c,m})$. Due to

$$p(\varphi_n^{c,m} | \mathbf{b}_n^{c,m}) \propto p(\mathbf{b}_n^{c,m} | \varphi_n^{c,m}) p(\varphi_n^{c,m}) \quad (11)$$

and

$$p(\mathbf{b}_n^{c,m} | \varphi_n^{c,m}) = \text{Bern}(\sigma(\varphi_n^{c,m})) = \frac{\exp(\varphi_n^{c,m} \mathbf{b}_n^{c,m})}{1 + \exp(\varphi_n^{c,m})} \quad (12)$$

we know that the estimation of $\varphi_n^{c,m}$ is determined based on the class label $\mathbf{b}_n^{c,m}$ (the parameters of the prior distribution $p(\varphi_n^{c,m})$ are quite small and have little effect on the estimation of $p(\varphi_n^{c,m} | \mathbf{b}_n^{c,m})$). In combination of (8), (10), (11), and (12), we know that the estimation of $\mathbf{w}_n^{c,m}$ is influenced by the class label $\mathbf{b}_n^{c,m}$. Similarly, $\mathbf{b}_n^{c,m}$ also influences the estimation of $\mathbf{z}_n^{c,m}$. Due to the distinctions of class labels,

the latent factor variables from different classes should have certain differences.

Based on the above analysis, we conclude that the introduction of class label contributes to the enhancement of the discriminability for latent factor variables from different classes, which helps to strengthen the separability of statistical models. As a result, our LA-FA is able to further improve the recognition performance under limited training samples.

Moreover, our LA-FA model first transforms the binary label $\mathbf{b}_n^{c,m}$ to the Gaussian distributed label preserving constraint term $\varphi_n^{c,m}$ by Sigmoid-Bernoulli nonlinear transformation, i.e., $\text{Bern}(\sigma(\cdot))$ which is referred to as the Sigmoid-Bernoulli hierarchy, and then projects $\varphi_n^{c,m}$ into the latent factor variable $\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}$ rather than directly mapping the label to the latent factor variable via a linear transformation, as shown in (6). To demonstrate the significance of the use of Sigmoid-Bernoulli hierarchy, we refer to the model that directly maps the label to the latent factor variable as the directly labeled FA (DL-FA) model. Thus, the DL-FA model can be represented as follows:

$$\begin{aligned} \mathbf{x}_n^{c,m} &= \mathbf{A}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}) + \mu^{c,m} + \varepsilon_n^{c,m} \\ \mathbf{b}_n^{c,m} &= \mathbf{D}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}) + \eta_n^{c,m}. \end{aligned} \quad (13)$$

Since the latent variable $\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m}$ is Gaussian distributed, the sample label $\mathbf{b}_n^{c,m}$ should also follow the Gaussian distribution if $\mathbf{b}_n^{c,m}$ is modeled via the approach in (13). However, the class label $\mathbf{b}_n^{c,m}$ is a discrete binary variable, and there is a mismatch of statistical characteristics between the class label and the statistical model in (13). To deal with this problem, the class label is described via the Bernoulli distribution, which is a good description for a binary vector composed of zero and one, in our LA-FA model. In addition, as the distribution parameter of Bernoulli, $\sigma(\varphi_n^{c,m})$ with $\varphi_n^{c,m}$ being Gaussian distributed has two advantages: 1) the Sigmoid function ensures the normalized Bernoulli distribution parameter, i.e., $\sigma(\varphi_n^{c,m}) \in (0, 1)$, which makes $\sigma(\varphi_n^{c,m})$ a probability value that can be used in Bernoulli distribution; 2) the Gaussian distributed label preserving constraint term can be better described through a linear transformation of Gaussian latent factor variable, which avoids the weakness of constraint performance caused by the mismatch between the binary label and Gaussian-based statistical model.

In summary, our LA-FA model introduces the label information in a more reasonable way to improve the separability of model, thus leading to the better recognition performance. The main advantages in our model structure come from two aspects as follows.

First, the LA-FA model introduces the class labels to restrain the learning of latent factor variables. By label constraint, the discrimination capability among the statistical models from different classes can be enhanced, which helps to achieve higher recognition accuracy with small training data size.

Second, our LA-FA model describes the generation of the discrete binary label via a nonlinear Sigmoid-Bernoulli

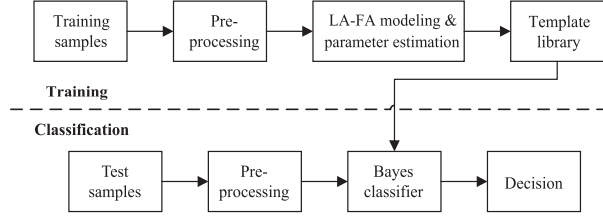


Fig. 6. Flowchart of the radar HRRP statistical recognition based on the LA-FA model.

hierarchy, i.e., $\text{Bern}(\sigma(\cdot))$, rather than a simple linear transformation, which improves the ability of our method to describe the class labels.

III. RADAR TARGET RECOGNITION BASED ON THE LA-FA MODEL

In our work, we build an LA-FA based recognition scheme for HRRP data, which is depicted in Fig. 6. The integrated scheme includes two stages, i.e., training stage and classification stage.

In the training stage, raw HRRP data is first preprocessed to deal with the sensitivity problems, such as target-aspect, time-shift, and amplitude-scale sensitivities. Next, a statistical model is learned for each frame of each target via our LA-FA model, and subsequently, the model parameter is estimated via Gibbs sampling algorithm, i.e., model learning. Finally, all the learned model parameters are stored as templates for Bayes classifier construction in the classification phase. In the classification stage, after data preprocessing for test HRRP samples, the Bayes classifier is constructed, with which the class membership of each test sample is determined based on the frame-conditional likelihoods of all frames from all classes. In detail, the frame-conditional likelihoods of a test sample given each target can be estimated according to the stored templates, and then, the maximum one in each target is selected as the corresponding class-conditional likelihood. Lastly, this test sample is assigned to the class with the maximum class-conditional likelihood.

In the following, we give a detailed introduction to the parameter estimation of the LA-FA model and provide an estimation method for the frame-conditional likelihood based on the LA-FA model.

A. Parameter Estimation With Data Augmentation

In the training stage, we seek to estimate the model parameters, i.e. $\Theta = \{\mathbf{a}_k, \mathbf{d}_k, \mathbf{w}_n^{c,m}, \mathbf{z}_n^{c,m}, \boldsymbol{\varphi}_n^{c,m}, \alpha^{c,m}, \gamma^{c,m}, \lambda^{c,m}, \mathbf{u}_k^{c,m}\}_{k=1, n=1, m=1, c=1}^{K, N, M_c, C}$. Our LA-FA model is a fully Bayesian treatment model, and the parameter calculation can be implemented via the Gibbs sampling-based Markov Chain Monte Carlo algorithm. In Algorithm 1, we summarize the process of LA-FA model learning based on the Gibbs sampling.

From Algorithm 1, we see that a Markov chain is constructed with the conditional posterior distributions. Based on the Bayes theorem, the posterior distribution of each parameter can be derived analytically. For

Algorithm 1: Parameter Estimation of the LA-FA Model Via Gibbs Sampling Algorithm.

Step 1: Initialize all the parameters in Θ and let $\tau = 1$, begin the Gibbs sampling procedure.

Step 2: Calculate the conditional posterior distributions of the LA-FA model (shown in Appendix) and construct a Markov chain with the posteriors.

Step 3: Burn-in: run the Markov chain to draw samples from the conditional posterior distributions, and then let $\tau = \tau + 1$.

Step 4: Return Step 3 if $\tau < I_0$; otherwise, continue Step 5. ($I_0 = 500$ in our experiment)

Step 5: Collect T_0 samples from the posterior distribution for all the parameters and calculate the mean values. (T_0 is set to 1000)

example, according to $p(\mathbf{w}_n^{c,m} | \mathbf{x}_n^{c,m}, \boldsymbol{\varphi}_n^{c,m}) \propto p(\mathbf{x}_n^{c,m} | \mathbf{w}_n^{c,m}) p(\boldsymbol{\varphi}_n^{c,m} | \mathbf{w}_n^{c,m}) p(\mathbf{w}_n^{c,m})$ given other model parameters, the posterior distribution of $\mathbf{w}_n^{c,m}$ (abbreviated as $p_{\mathbf{w}_n^{c,m}}$) is

$$p_{\mathbf{w}_n^{c,m}} = N(\mathbf{w}_n^{c,m} | \boldsymbol{\zeta}_{\mathbf{w}_n^{c,m}}, \boldsymbol{\Sigma}_{\mathbf{w}_n^{c,m}}) \quad (14)$$

where

$$\begin{aligned} \boldsymbol{\zeta}_{\mathbf{w}_n^{c,m}} &= \boldsymbol{\Sigma}_{\mathbf{w}_n^{c,m}} \left[\gamma^{c,m} \text{diag}(\mathbf{z}_n^{c,m}) \mathbf{A}^T \left(\mathbf{x}_n^{c,m} - \boldsymbol{\mu}^{c,m} \right. \right. \\ &\quad \left. \left. - \sum_{i=1, i \neq k}^K \mathbf{a}_i w_{n,i}^{c,m} z_{n,i}^{c,m} \right) + \lambda^{c,m} \text{diag}(\mathbf{z}_n^{c,m}) \right. \\ &\quad \left. \times \mathbf{D}^T \left(\boldsymbol{\varphi}_n^{c,m} - \sum_{i=1, i \neq k}^K \mathbf{d}_i w_{n,i}^{c,m} z_{n,i}^{c,m} \right) \right] \end{aligned}$$

and

$$\begin{aligned} \boldsymbol{\Sigma}_{\mathbf{w}_n^{c,m}} &= [\gamma^{c,m} \text{diag}(\mathbf{z}_n^{c,m}) \mathbf{A}^T \mathbf{A} \text{diag}(\mathbf{z}_n^{c,m}) \\ &\quad + \lambda^{c,m} \text{diag}(\mathbf{z}_n^{c,m}) \mathbf{D}^T \mathbf{D} \text{diag}(\mathbf{z}_n^{c,m}) + \alpha^{c,m} \mathbf{I}_K]^{-1}. \end{aligned}$$

Apparently, the latent factor variable $\mathbf{w}_n^{c,m}$ is not only related to the observation $\mathbf{x}_n^{c,m}$, but also is adjusted by the label preserving constraint term $\boldsymbol{\varphi}_n^{c,m}$. Since $\boldsymbol{\varphi}_n^{c,m}$ preserves the property of class label, the label information has the effect on the estimation of $\mathbf{w}_n^{c,m}$. Therefore, the differences of class labels will enhance the distinctions of latent factor variables, and furthermore, strengthen the separability of statistical models from different classes, which is beneficial to the improvement of recognition performance.

In addition, the class label $\mathbf{b}_n^{c,m}$ is connected to the model via the nonlinear Sigmoid-Bernoulli hierarchy $\text{Bern}(\sigma(\cdot))$, which leads to the posterior distribution of $\boldsymbol{\varphi}_n^{c,m}$ intractable. To deal with this problem, an augmented variable $\boldsymbol{\delta}_n^{c,m} = [\delta_{n,1}^{c,m}, \dots, \delta_{n,L}^{c,m}]^T$ with $p(\delta_{n,l}^{c,m}) = \text{PG}(1, 0)$ is introduced. Here $\text{PG}(\cdot)$ denotes Poly-Gamma distribution. Then, the likelihood $p(\mathbf{b}_{n,l}^{c,m} | \varphi_{n,l}^{c,m})$ can be regarded as the marginal

from a joint distribution $p(b_{n,l}^{c,m}, \delta_{n,l}^{c,m} | \varphi_{n,l}^{c,m})$, which has the explicit expression as follows:

$$p(b_{n,l}^{c,m}, \delta_{n,l}^{c,m} | \varphi_{n,l}^{c,m}) = 2^{-1} \exp((b_{n,l}^{c,m} - 1/2) \varphi_{n,l}^{c,m}) \times \exp(-\delta_{n,l}^{c,m} \varphi_{n,l}^{c,m^2}/2) \text{PG}(1, 0). \quad (15)$$

According to the Bayes theorem $p(\varphi_{n,l}^{c,m} | b_{n,l}^{c,m}, \delta_{n,l}^{c,m}) \propto p(b_{n,l}^{c,m}, \delta_{n,l}^{c,m} | \varphi_{n,l}^{c,m}) p(\varphi_{n,l}^{c,m})$, a Gaussian-based posterior distribution can be derived for $\varphi_{n,l}^{c,m}$, and the details are shown in the Appendix. Consequently, the Gibbs sampling algorithm can be implemented to repeatedly sample each parameter from its conditional posterior distribution.

B. Noise-Robust Recognition via LA-FA-Based Bayes Classifier

After learning the LA-FA model for each target c with $c = 1, \dots, C$, the sampling means of the model parameters are obtained. If a test sample $\mathbf{x}^*$ is measured, the frame-conditional likelihood of $\mathbf{x}^*$ given the frame m of target c is

$$p(\mathbf{x}^* | m, c) = N(\boldsymbol{\mu}^{c,m}, \boldsymbol{\Sigma}^{c,m}) \quad (16)$$

with

$$\begin{aligned} \boldsymbol{\Sigma}^{c,m} &= \langle \mathbf{A} \rangle \boldsymbol{\Omega}^{c,m} \langle \mathbf{A} \rangle^T + \langle \gamma^{c,m} \rangle^{-1} \mathbf{I}_P \\ \boldsymbol{\Omega}^{c,m} &= \frac{1}{N} \sum_{n=1}^N ((\mathbf{w}_n^{c,m}) \odot (\mathbf{z}_n^{c,m})) ((\mathbf{w}_n^{c,m}) \odot (\mathbf{z}_n^{c,m}))^T \end{aligned} \quad (17)$$

where $\langle \cdot \rangle$ represents the sampling mean of the variable.

Since the frame models from a given target are assumed to be independent, the class-conditional likelihood of $\mathbf{x}^*$ given target c is $p(\mathbf{x}^* | c) = \max_m p(\mathbf{x}^* | m, c)$ with $m = 1, \dots, M_c$. According to the Bayes theorem $p(c | \mathbf{x}^*) \propto p(\mathbf{x}^* | c) p(c)$, the class membership of $\mathbf{x}^*$ can be assigned to the class with the maximum posterior probability, i.e., $c^* = \text{argmax}_c p(c | \mathbf{x}^*)$.

In the above discussion, we assume the noise level of the test sample is same as those of the training samples. However, in the real application, the training data are usually collected under the condition of high SNR via some cooperative measurement experiments, whereas the test samples are usually obtained in the noncooperative circumstance, where the high SNR cannot be guaranteed. Since the noise component can change the statistical distribution of the noisy test HRRP sample, the recognition performance degrades dramatically for the test HRRP sample collected under low SNR and discriminated with the statistical model trained under high SNR. Motivated by this, we propose a noise-robust modification method for Gaussian-based statistical models.

In the hypothesis of Gaussian distribution, the measured test HRRP sample under low SNR can be expressed as $\tilde{\mathbf{x}}^* \approx \mathbf{x}^* + \mathbf{n}^*$, where $\mathbf{x}^*$ represents the original HRRP sample with high SNR and $\mathbf{n}^*$ represents the Gaussian noise variable introduced from the test condition. Then, the frame-conditional likelihood of $\mathbf{x}^*$ given frame m of target c can be expressed as follows:

$$p(\tilde{\mathbf{x}}^* | c, m) = N(\tilde{\boldsymbol{\mu}}^{c,m}, \tilde{\boldsymbol{\Sigma}}^{c,m}) \quad (18)$$

TABLE II
Parameters of Three Airplanes and Radar in the Experiment

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

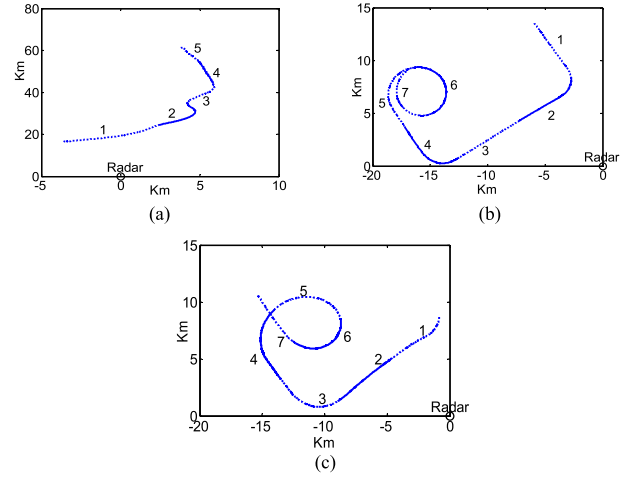


Fig. 7. Projections of target trajectories onto the ground plane.
(a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

with

$$\tilde{\boldsymbol{\mu}}^{c,m} \approx \boldsymbol{\mu}^{c,m}, \tilde{\boldsymbol{\Sigma}}^{c,m} \approx \boldsymbol{\Sigma}^{c,m} + \sigma_n^2 \mathbf{I}_P \quad (19)$$

where $\boldsymbol{\mu}^{c,m}$ is the statistical mean of training HRRPs from the m th frame of target c , $\boldsymbol{\Sigma}^{c,m}$ is the covariance of the m th frame for target c and is estimated via the LA-FA model learned in the training stage; σ_n^2 is the noise power of the test sample and can be easily estimated via some preprocessing method [5].

From (18) and (19), we know that the covariance matrix estimated via the training HRRP samples with high SNR is modified according to the noise power of test HRRPs. By this noise-robust modification method, the noise level of the statistical model can better match that of the test sample, thus can potentially obtain higher recognition accuracy under the test condition of low SNR.

IV. EXPERIMENTAL RESULTS

A. Data Description

The measured HRRP data used in this paper is the ISAR data from three real airplanes (Yark-42, Cessna Citation S/II, and An-26) with the center frequency of 5520 MHz and the bandwidth of 400 MHz. These data are also used in [3]–[7], [12], [21], [26], [29]. The parameters of the airplane targets and measurement radar are shown in Table II.

The projections of target trajectories onto the ground plane are shown in Fig. 7, from which the aspect angle of an airplane can be estimated according to its relative

position to radar. As shown in Fig. 7, the measured data are segmented, where each segment contains 25 600 HRRP samples and each sample is a 256-length vector, i.e., $P = 256$. In our experiments, we take samples from the second segment and a part of the fifth segment of Yark-42, the sixth and seventh segments of Cessna Citation S/II, the fifth and sixth segments of An-26 as the training data, and take 5200 samples from other segments as the test data.

It is a prerequisite for radar HRRP target recognition to deal with the target-aspect, time-shift, and amplitude-scale sensitivity problems. As discussed in the Introduction, we use the aspect-frame partition method [4] to tackle the target-aspect sensitivity of HRRP training dataset. In detail, for each frame of each target, the change of the target-aspect is equal to the critical angle corresponding to the nonoccurrence of scatterers' MTRC. According to this, there are 35 aspect-frames for Yark-42, 50 for Cessna Citation S/II, and 50 for An-26, of which each aspect-frame totally has 1024 samples, i.e., $C = 3$, $M_1 = 35$, $M_2 = M_3 = 50$, and $N = 1024$. Since an HRRP is only a part of received radar echo extracted by a range window, where target signal is included, the position of the target signal in an HRRP can vary with the measurement. This is the time-shift sensitivity of HRRP [25]. Here, we use a traditional motion compensation method, i.e., centroid alignment [15], to tackle the time-shift sensitivity. In addition, since the intensity of an HRRP is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses, and so on, HRRPs measured by different radars or under different conditions will have different amplitude-scales [6]. To deal with this amplitude-scale sensitivity, each HRRP sample is normalized by dividing its L_2 norm, $\mathbf{x}_n = \mathbf{x}_n / \|\mathbf{x}_n\|_2$.

In our experiments, eight training datasets are considered to train the LA-FA model: 10×135 , 20×135 , 30×135 , 40×135 , 50×135 , 70×135 , 90×135 , and 120×135 , where 10×135 means that 10 HRRP training samples are randomly selected from each of the 135 aspect-frames, similarly for other data size expressions. Moreover, the selecting work is repeated 20 times, and the results are averaged over the 20 random selections. Meanwhile, the standard deviations are also calculated and presented by error bars to describe the uncertainty caused by random selections.

In addition, the HRRPs used in our experiment are of different SNRs and measured under high SNR ($\text{SNR} \geq 30$ dB). To evaluate the robustness of our noise-robust modification method, we add simulated white noises to the test data to obtain noisy HRRP test samples. Here, the SNR is defined as the ratio of the average power of signal to that of the noise for each sample. In detail, the SNR is expressed as follows:

$$\text{SNR} = 10 \times \log_{10} \left(\frac{\bar{P}_x}{P_{\text{Noise}}} \right) = 10 \times \log_{10} \left(\frac{\sum_{p=1}^P P_{x_p}}{P \times P_{\text{Noise}}} \right) \quad (20)$$

where $\bar{P}_x$ denotes the average power of the original sample, P_{x_p} denotes the energy of the p th range cell in the original

sample, P denotes the dimension of the sample, and P_{Noise} denotes the power of noise. Thus, $10^{\text{SNR}/10}$ equals to the ratio of the average power of original sample to the power of the noise. From (20), we know that L_2 normalization of HRRP does not change the SNR and all HRRPs are still of different SNRs. However, for HRRP-based radar target recognition, the training data are obtained under the condition of high SNR, and the noise has little effect on the statistical characteristics of training samples. Accordingly, in the training stage, we ignore the impact of different SNRs among HRRPs on the learning of models. For the test HRRP samples, their predictions are uncorrelated. Therefore, the same SNR for test samples is not required.

B. Influence of the Hyperparameters

As the prior knowledge, the hyperparameters greatly affect the performance of the model. Due to the posterior of the model parameter is related to the likelihood function and the prior distribution, the unreasonable settings of hyperparameters will decrease the parameter estimation precision. In the following, we give an introduction to the settings of hyperparameters in the LA-FA model.

There are two kinds of hyperparameters: the hyperparameters in Gamma distribution including $a_0, b_0, e_0, f_0, g_0, h_0$, and the hyperparameters in Beta distribution including c_0, d_0 . For the hyperparameters in Gamma distribution, we usually assign small values to them to reduce the influence of priors on the estimations of model parameters. That is, the parameter estimations (the posteriors) are entirely up to the observed data (the likelihoods). This avoids the inaccurate parameter estimations caused by unreasonable hyperparameter settings. Therefore, in our paper, a_0, b_0, e_0, f_0, g_0 , and h_0 are set to 10^{-6} , same as the corresponding prior settings in [5].

For the hyperparameters in Beta distribution, i.e., c_0, d_0 , they control the sparsity of $\mathbf{z}_n^{c,m}$. The expectation of $z_{n,k}^{c,m}$ ($k = 1, \dots, K$) is $c_0/(c_0 + d_0)$. The smaller the $c_0/(c_0 + d_0)$ is, the smaller the probability of $z_{n,k}^{c,m}$ being one is, and the sparser the $\mathbf{z}_n^{c,m}$ is. Since only a small subset from dictionary atoms in factor dictionary is used to describe $\mathbf{x}_n^{c,m}$, $\mathbf{z}_n^{c,m}$ should be a sparse vector, and thus $c_0/(c_0 + d_0)$ should be a small value. Moreover, for the LA-FA model with different training samples per frame, the number of selected dictionary atoms is often different, i.e., different sparsity. Therefore, $c_0/(c_0 + d_0)$ should also vary with the training sample number per frame. Based on the analysis above, we set c_0 to $1/K$ with K being the total number of dictionary atoms ($K = 256$ in our experiment) and d_0 to $\rho(K - 1)/K$ with ρ being a tunable parameter.

Fig. 8 shows the recognition accuracies of the LA-FA model, versus ρ , under different training sample size conditions. From each subfigure in Fig. 8, we note that the hyperparameter d_0 affects the model performance. The values of ρ with optimal recognition rates (denoted as ρ_{opt}) are 50, 40, and 20, respectively. Thus, the optimal d_0 ($\rho_{\text{opt}}(K - 1)/K$) is much larger than c_0 ($1/K$), and the expectation of $z_{n,k}^{c,m}$, i.e., $c_0/(c_0 + d_0)$, is a much small

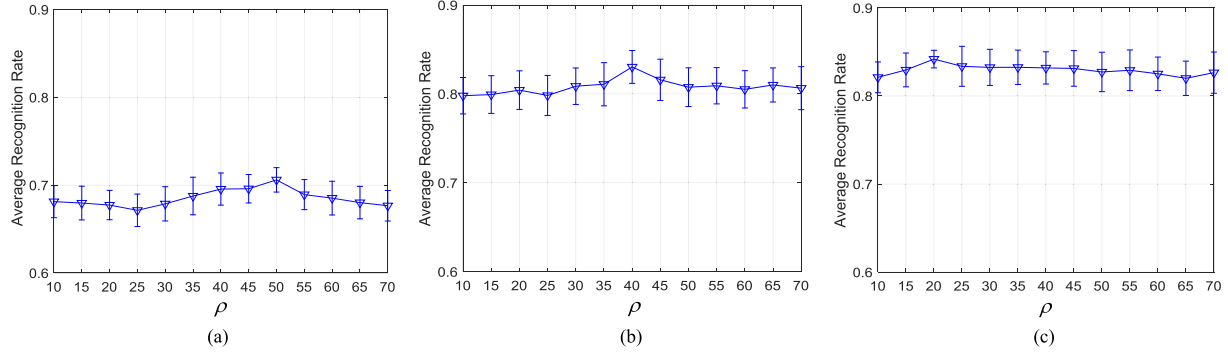


Fig. 8. Variation of recognition rate with ρ , via the LA-FA model with different training data sizes. (a) About 10 training samples per frame. (b) About 50 training samples per frame. (c) About 120 training samples per frame.

value for each subfigure. Via such hyperparameters in Beta distribution, most elements in the inferred posterior of $\mathbf{z}_n^{c,m}$ will be zero, which can guarantee the sparsity of $\mathbf{z}_n^{c,m}$. In addition, by comparison among subfigures in Fig. 8, we can find that the optimal settings of d_0 are different for the LA-FA model with different training data sizes. The more training samples per frame, the smaller d_0 the optimal recognition rate corresponds to. This is because when there are only a few training samples, the required number of dictionary atoms for the description of observations is relatively small, thus d_0 being a large value. With the increase of training data, it needs more dictionary atoms to describe observed data, and the sparsity parameter d_0 should be relatively small. Therefore, we should adjust the hyperparameter d_0 to obtain the better recognition performance in the model learning.

C. Training Result Analysis

In our LA-FA model, HRRP samples from all aspect-frames of all targets share a factor dictionary $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_K]$ with large size, and the subdictionary for each frame of each target, i.e., $\{\mathbf{A}^{c,m}\}_{m=1, c=1}^{M_c, C}$, is formed by selecting only a small subset from the dictionary atoms in $\mathbf{A}$. In detail, which dictionary atoms to compose $\mathbf{A}^{c,m}$ for observations $\{\mathbf{x}_n^{c,m}\}_{n=1}^N$ from frame m of target c is automatically determined via $\mathbf{z}_n^{c,m} = [z_{n,1}^{c,m}, \dots, z_{n,K}^{c,m}]^T$ ($n = 1, \dots, N$). If there are nonzero elements in $\{z_{n,k}^{c,m}\}_{n=1}^N$, the k th dictionary atom $\mathbf{a}_k$ is selected for composing $\mathbf{A}^{c,m}$; otherwise, $\mathbf{a}_k$ is not utilized. For different frames, the learned selection vectors are different, and thus, the subdictionary of each frame is also different.

In our experiment, we set the number of shared dictionary atoms to 256, i.e., $K = 256$. The initial values of the shared dictionary $\mathbf{A}$ and selection vectors $\{\mathbf{z}_n^{c,m}\}_{n=1, m=1, c=1}^{N, M_c, C}$ are sampled from distributions $N(\mathbf{0}, 1/P\mathbf{I}_p)$ and $\text{Bern}(1)$, respectively, and their final values are obtained via the Gibbs sampling iteration algorithm. In addition, due to $z_{n,k}^{c,m} \sim \text{Bern}(u_k^{c,m})$, the learned $\mathbf{u}^{c,m}(\mathbf{u}^{c,m} = [u_1^{c,m}, \dots, u_K^{c,m}]^T, m = 1, \dots, M_c, c = 1, \dots, C)$ can reflect the usage of dictionary atoms for each frame. The larger the $u_k^{c,m}$ is, the more likely the $\{z_{n,k}^{c,m}\}_{n=1}^N$ are to be

one, and the higher usage the dictionary atom $\mathbf{a}_k$ has for frame m of target c . To demonstrate the usage of dictionary atoms for each frame, we show the binary image of sampling mean of $\mathbf{u}^{c,m}$, i.e., $\langle \mathbf{u}^{c,m} \rangle = [\langle u_1^{c,m} \rangle, \dots, \langle u_K^{c,m} \rangle]^T$ with $\langle \cdot \rangle$ denoting the sampling mean of a variable, from 5 frames of each target, via the LA-FA model with 50 training samples per frame in Fig. 9. For each target c , if $\langle u_k^{c,m} \rangle$ is above 0.5, we consider the dictionary atom k suitable for frame m , then replace the value of $\langle u_k^{c,m} \rangle$ with one and show it with the black line in Fig. 9. If $\langle u_k^{c,m} \rangle$ is below 0.5, we consider that the dictionary atom k is not used for frame m , then replace the value of $\langle u_k^{c,m} \rangle$ with zero and show it with the white line Fig. 9.

As we can see from Fig. 9, there are small amount of black lines for each frame, i.e., the learned $\mathbf{u}^{c,m}$ is sparse, which demonstrates that the initial number of dictionary atoms ($K = 256$) is sufficient enough to represent the observed HRRPs from all frames of all targets. Meanwhile, for different frames, the dictionary atoms used to describe the observations are different. Therefore, the binary selection vector can automatically select the suitable dictionary atoms for the corresponding aspect-frame HRRP data. In addition, the number of dictionary atoms varies across the frame index. By contrast, we fix the same dictionary atom number for all frames in the model without selection vector $\mathbf{z}_n^{c,m}$, which may lead to overfitting or underfitting for some frames. Moreover, although the selected dictionary atoms for each frame are not exactly the same, some different frames may select the same dictionary atoms, thus confirming the rationality of sharing factor dictionary across all frames of all targets.

In the LA-FA model, the class label is first transformed to the label preserving constraint term via the Sigmoid-Bernoulli hierarchy, and then the label preserving constraint term is used to restrain the learning of latent factor variable. To demonstrate that the label preserving constraint terms among different classes can preserve the differences of class labels, we show the gray images of the learned label preserving constraint terms from the two frames of each target, via the LA-FA model with 50 training samples per frame, i.e., $\{\varphi_n^{c,m}\}_{n=1}^{50}$ ($m = 1, 2; c = 1, 2, 3$), in Fig. 10. In the experiment, the class labels of Yark-42, Cessna Citation S/II

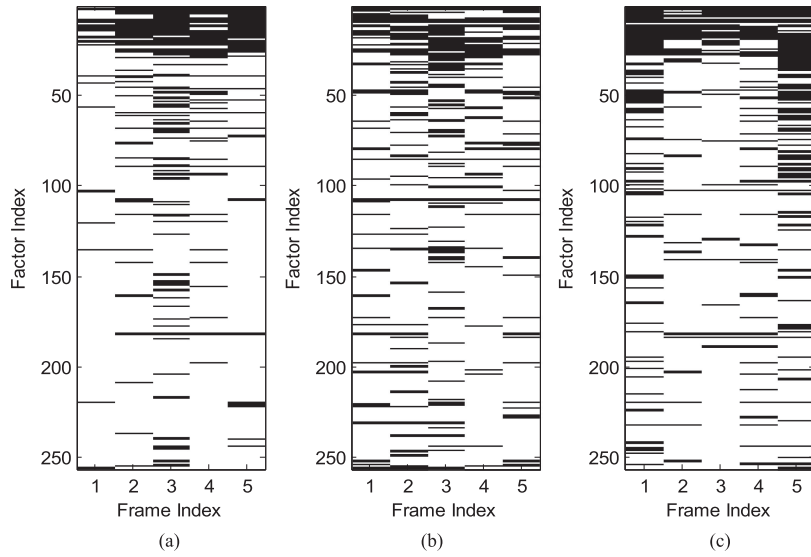


Fig. 9. Sampling mean of the learned $\mathbf{u}^{c,m}$ from five frames of each target, via the LA-FA model with 50 training samples per frame. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

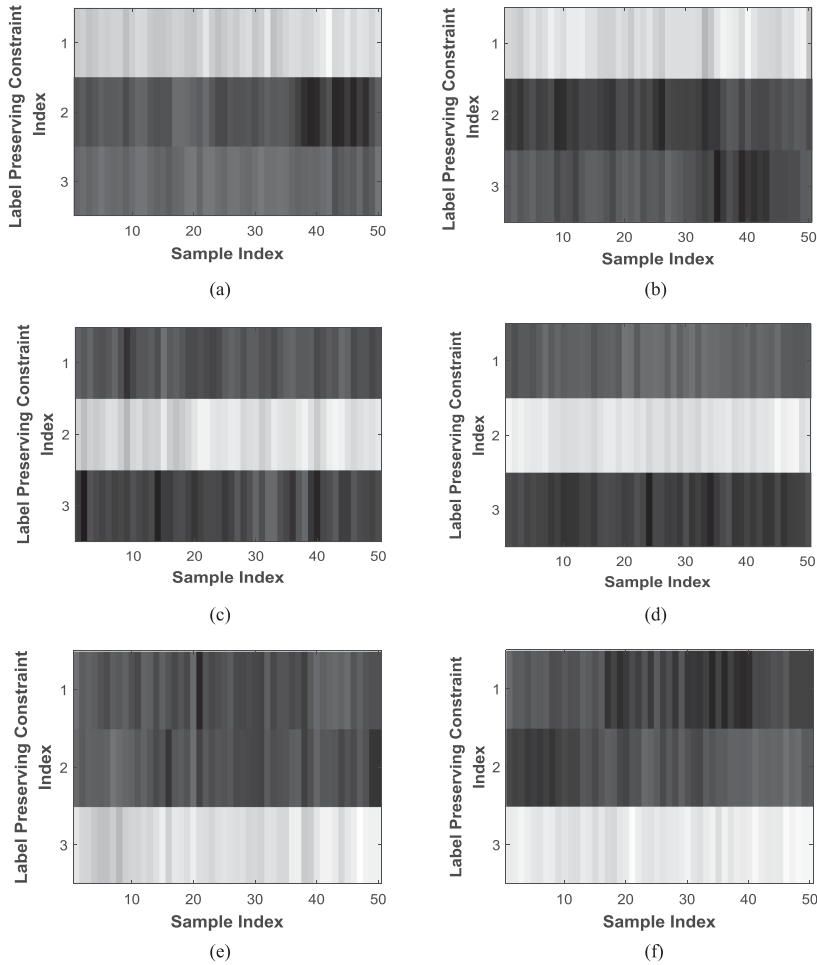


Fig. 10. Gray images of label preserving constraint terms from two frames of each target, via the LA-FA model with 50 training samples per frame. (a) and (b) correspond to the first and second frames of Yark-42, respectively, with label being $[1, 0, 0]^T$. (c) and (d) correspond to the first and second frames of Cessna Citation S/II, respectively, with label being $[0, 1, 0]^T$. (e) and (f) correspond to the first and second frames of Cessna Citation S/II, respectively, with label being $[0, 0, 1]^T$.

TABLE III
 R_{KL} of LA-FA, MTL-FA, and DL-FA Models Under Different Training Data Sizes

Training data number per frame	10	20	30	40	50	70	90	120
LA-FA model	1.31 (0.03)	1.49 (0.04)	1.53 (0.04)	1.59 (0.05)	1.65 (0.05)	1.67 (0.04)	1.71 (0.05)	1.63 (0.05)
MTL-FA model	1.27 (0.03)	1.29 (0.03)	1.48 (0.04)	1.44 (0.04)	1.45 (0.05)	1.49 (0.03)	1.51 (0.03)	1.56 (0.04)
DL-FA model	1.29 (0.03)	1.47 (0.04)	1.51 (0.04)	1.55 (0.04)	1.59 (0.04)	1.63 (0.04)	1.56 (0.03)	1.57 (0.03)

and An-26 are set to $[1, 0, 0]^T$, $[0, 1, 0]^T$, and $[0, 0, 1]^T$, respectively. In addition, the larger the element in $\phi_n^{c,m}$ is, the paler the corresponding color is in Fig. 10.

From Fig. 10(a) and (b), we see that the color of the first line is paler than those of the other two lines in the gray images of Yark-42, which demonstrates that the first elements of all the learned label preserving constraint terms are larger than the other two elements. This relationship among the elements in the label preserving constraint term is consistent with that in the class label of Yark-42, i.e., $[1, 0, 0]^T$. That is, the larger value in label preserving constraint term corresponds to the nonzero element in sample label; on the contrary, the smaller value corresponds to the zero element, similarly for gray images of Cessna Citation S/II and An-26. Based on the above analysis, we conclude that there exists a corresponding relationship between the sample label and label preserving constraint term, and the label information can be preserved in the label preserving constraint term. Therefore, the label preserving constraint term can be used to restrict the learning of model parameters, and furthermore, improve the separability of statistical models from different targets.

The joint-likelihood of observations and sample labels can reflect the descriptive performance of model to observed data and label. The greater the likelihood is, the stronger descriptive ability the model has. To validate the effectiveness of Sigmoid-Bernoulli hierarchy on the description of binary class labels, we present the means of joint log-likelihoods, i.e., $(\sum_{c=1}^C M_c \cdot N)^{-1} \sum_{c=1, m=1, n=1}^{C, M_c, N} \ln p(\mathbf{x}_n^{c,m}, \mathbf{b}_n^{c,m})$, via the LA-FA and DL-FA models with different training data sizes in Fig. 11. As we can see from Fig. 11, the means of log-likelihoods of the LA-FA model are larger than those of the DL-FA model, which indicates that our LA-FA model can more accurately describe the observed data and their class labels via Sigmoid-Bernoulli hierarchy, thus avoiding the weakness of label constraint performance caused by the inaccurate label description.

The Kullback–Leibler (KL) distance is regularly used to measure the similarity between two statistical models in pattern recognition. The smaller the KL distance is, the higher similarity the statistical models have. Here, we define an average KL distance rate R_{KL} as follows:

$$R_{KL} = \text{mean} \left(\sum_{\text{diff}} \text{KL}_{\text{diff}} \right) / \text{mean} \left(\sum_{\text{same}} \text{KL}_{\text{same}} \right) \quad (21)$$

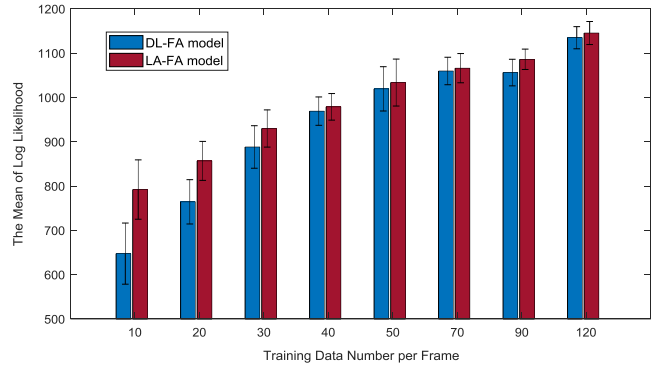


Fig. 11. Histograms of the means of log likelihoods for DL-FA and LA-FA models with different number of training samples.

where $\text{mean}(\cdot)$ denotes calculation of the mean value, KL_{diff} is the KL distance between two statistical models from different classes, the KL_{same} denotes the KL distance between two statistical models from the same class, $\Sigma_{\text{diff}}(\cdot)$ and $\Sigma_{\text{same}}(\cdot)$ denote the summations of KL_{diff} from different classes and KL_{same} from the same class, respectively.

Table III shows the R_{KL} of the LA-FA, DL-FA, and MTL-FA models with different training data sizes, where the number in the parentheses represents the standard deviation. From Table III, we can see the R_{KL} of LA-FA model is larger than those of the other two models, and the R_{KL} of each model increases with the increase of training sample number. Thus, the class labels enhance the separability of the statistical models for different classes, and the preserving constraint term via a nonlinear Sigmoid-Bernoulli hierarchy is more appropriate than introducing class labels directly.

D. Recognition Performance

In the following, we will evaluate the recognition performance of our LA-FA model by comparing with the principal component analysis based subspace approximation (referred to as PCA) model [4], PPCA model [4], STL-FA model, MTL-FA model, and DL-FA model. The recognition rates of our LA-FA model and other related models, versus the size of training data are shown in Fig. 12.

Fig. 12 indicates that the average recognition rate of each model increases with the increase of training sample number. The reason is that when the number of training

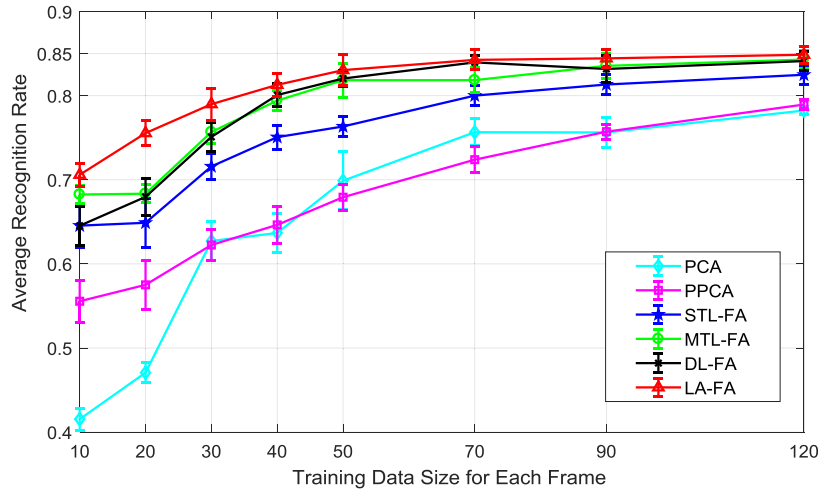


Fig. 12. Variations of average recognition rate with different training data sizes.

samples is large, the model parameter estimation becomes much accurate. Thus, the recognition rate is high. For the PCA and PPCA models, they both need to estimate $P \times P$ dimensional sample covariance matrix for each frame of each target. Generally, an HRRP sample includes many range cells ($P = 256$ in this paper); therefore, the computation burdens of parameter estimations for these two models are extremely large. However, the STL-FA model does not directly estimate the sample covariance matrix, but estimates the factor dictionary and covariance matrix of the noise variable. Then, the number of model parameters in the STL-FA model is smaller than those in the PCA and PPCA models. Therefore, the STL-FA model has the higher average recognition rates than the PCA model and PPCA model under limited training samples, as shown in Fig. 12. Moreover, the recognition performance of MTL-based models (MTL-FA model, DL-FA model, and our LA-FA model) is better than that of the STL-FA model, due to their fewer model parameters to be estimated by sharing factor dictionary. In addition, the learning of model parameters in the DL-FA model and LA-FA model is constrained by class labels, which enhances the separability of statistical models from different classes. Thus, they have higher recognition rates than the MTL-FA model. Meanwhile, since our LA-FA model introduces the class labels in a more reasonable way than the DL-FA model, the average recognition rates of LA-FA model are higher than those of DL-FA model. If we assume that the average recognition rate threshold is 70%, our LA-FA model can work even when the training data size per frame is smaller than 20, in which case the other related models have lost the recognition capability. In summary, our LA-FA model is more suitable for the recognition problem with small training size.

The above recognition results are obtained under the condition that both the training and test HRRP data have high SNR ($\text{SNR} \geq 30$ dB). However, in the real application, the high SNR of test samples cannot be guaranteed. When the test HRRP samples are of low SNR, there exists mismatch of statistical characteristics between the test data

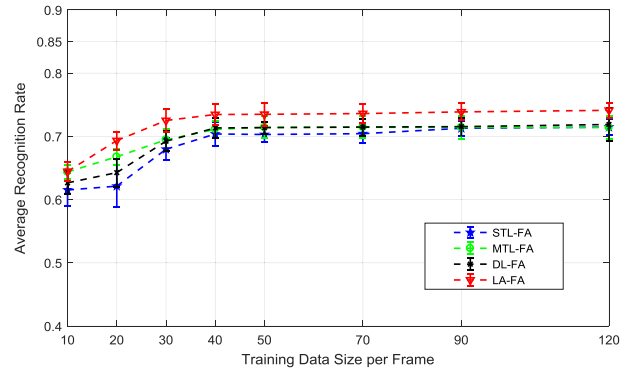


Fig. 13. Variations of average recognition rate with different training data sizes, under the test condition of $\text{SNR} = 15$ dB.

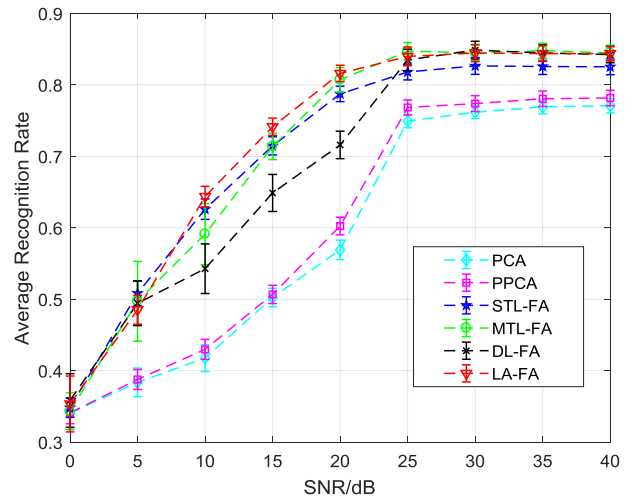


Fig. 14. Variation of recognition performance with SNR, via six statistical models under the condition of 120 training samples per frame.

and statistical model trained under high SNR. In Fig. 13, we show the variation of recognition results with training data size, under the test condition of $\text{SNR} = 15$ dB. Here it should be noted that the recognition curves of PCA and

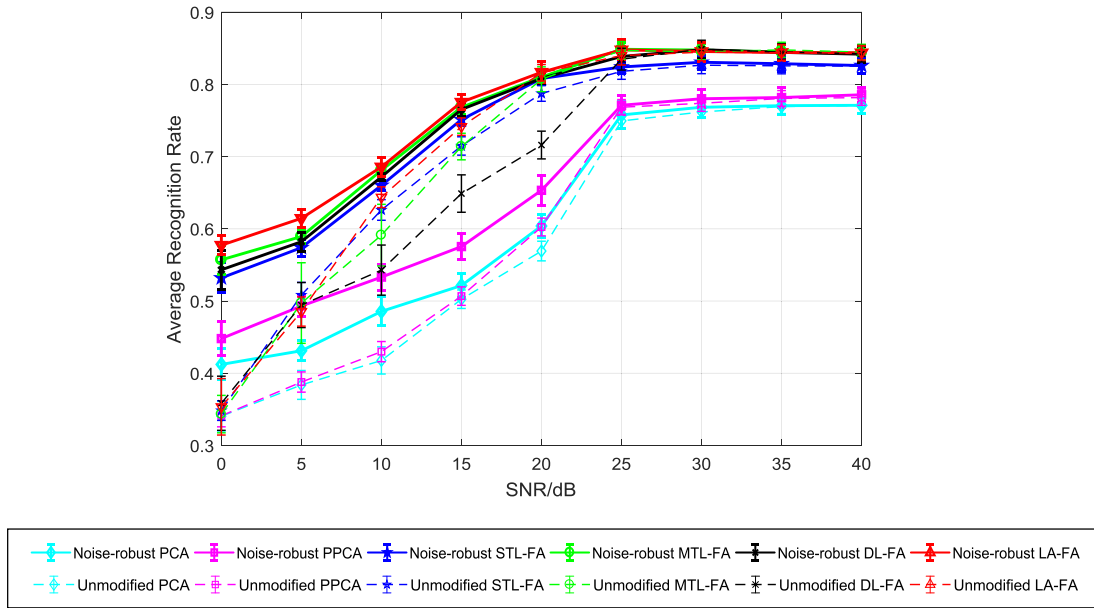


Fig. 15. Comparison of the recognition performance under different test SNRs, via the modified and unmodified models with 120 training samples per frame.

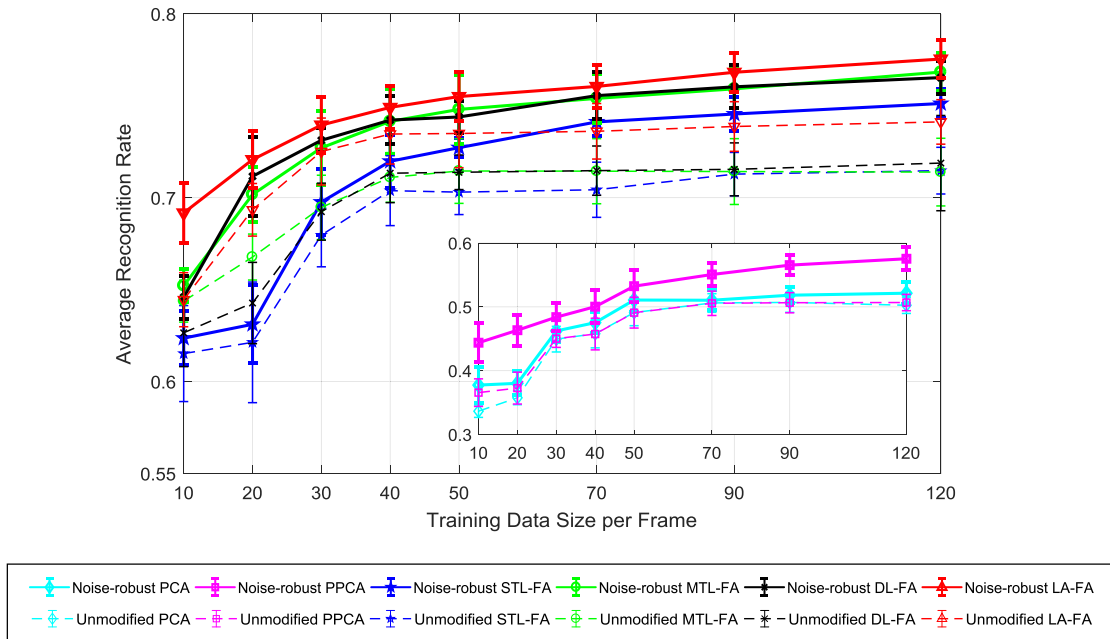


Fig. 16. Variation of recognition performance with training data size, under the test condition of SNR = 15 dB.

PPCA are omitted due to their recognition rates lower than 30%. By comparing Fig. 12 with Fig. 13, we see that the recognition accuracies of all models decrease sharply when the test data are of low SNR, and even our LA-FA model loses the recognition capability in the case of training sample number per frame less than 30.

Fig. 14 depicts recognition performance under different SNRs, via six models with 120 training samples for each frame. As we see in Fig. 14, noise has a small influence on the statistical distribution of test HRRP sample under high SNR conditions ($\text{SNR} \geq 25$ dB), in which case the

recognition accuracies of models are basically unchanged. When $\text{SNR} \leq 20$ dB, the recognition performance dramatically worsens, and the lower the test SNR is, the lower the recognition rate is.

From Figs. 13 and 14, we can conclude that the SNR of test data greatly influences the performance of recognition system. The mismatch between the learned statistical model and test samples caused by test noise should also be considered when we apply a statistical model to the HRRP-based radar target recognition. Based on this demand, we propose a noise-robust modification method to improve the

recognition performance of statistical models, under the test condition of low SNR.

In the following, we apply the noise-robust modification method to above Gaussian statistical models (LA-FA, DL-FA, MTL-FA, STL-FA, PPCA, and PCA models) and give a comparison of recognition performance under different test SNR conditions, via the modified and unmodified models with 120 training samples per frame, in Fig. 15. In addition, we refer to the model with noise-robust modification as the noise-robust model and the model without modification as the unmodified model.

As shown Fig. 15, the recognition rates of the noise-robust statistical models are better than those of models without modification, especially when the SNR of the test data is lower than 20 dB, which demonstrates the superiority of our noise-robust modification method. In Fig. 16, we present the variation of recognition performance with training data size, under the test condition of SNR = 15 dB. Here the recognition rates of the PCA and PPCA models are shown in subfigure for clarity. From Fig. 16, we also see the better recognition performance of the statistical models with noise-robust modification. In particular, our noise-robust LA-FA model has the ability to recognize the target when there are 20 training samples for each frame.

V. CONCLUSION

As a further investigation of the recent work [5] on radar HRRP data statistical modeling and recognition, this paper develops an LA-FA model to achieve better recognition performance. The LA-FA model incorporates label information into the MTL-FA model via a Sigmoid-Bernoulli hierarchy and uses the differences of class labels to enhance the interclass distinguishability of statistical models learned by the MTL-FA model. Consequently, the LA-FA model has the stronger recognition ability than the MTL-FA model, especially when there are limited training samples. The recognition results of the proposed model with different training data sizes demonstrate that our LA-FA model is more applicable to small sample target recognition.

Considering the mismatch of SNR between training data and test data in the real application, we also introduce a noise-robust modification method, which updates the noise level parameter in the statistical model in the classification phase to match that of the received test sample. Therefore, the proposed method can significantly improve the performance of radar HRRP recognition under the test condition of low SNR. The recognition experiments for noisy test HRRP data validate the effectiveness of our noise-robust modification method.

APPENDIX POSTERIOR DISTRIBUTIONS OF PARAMETERS IN THE LA-FA MODEL

In this section, we would like to give the details about the derivations of the Gibbs update equations for the LA-FA model. In Gibbs sampling algorithm, we seek to obtain the posterior distributions of all parameters $\Theta = \{\mathbf{a}_k, \mathbf{d}_k, \mathbf{w}_n^{c,m},$

$\mathbf{z}_n^{c,m}, \boldsymbol{\varphi}_n^{c,m}, \boldsymbol{\alpha}^{c,m}, \boldsymbol{\gamma}^{c,m}, \boldsymbol{\lambda}^{c,m}, \mathbf{u}_k^{c,m}\}_{k=1, n=1, m=1, c=1}^{K, N, M_c, C}$, given the observations $\mathbf{X} = \{\mathbf{x}_n^{c,m}\}_{n=1, m=1, c=1}^{N, M_c, C}$, class labels $\mathbf{B} = \{\mathbf{b}_n^{c,m}\}_n = 1, m = 1, c = 1^{N, M_c, C}$, and the hyperparameters $\Upsilon = \{a_0, b_0, c_0, d_0, e_0, f_0, g_0, h_0\}$.

The full posterior of parameters $p(\Theta|\mathbf{X}, \mathbf{B}, \Upsilon)$ can be expressed as follows:

$$\begin{aligned} p(\Theta|\mathbf{X}, \mathbf{B}, \Upsilon) &\propto p(\mathbf{X}|\Theta, \Upsilon) p(\mathbf{B}|\Theta, \Upsilon) p(\Theta|\Upsilon) \\ &= \prod_{c=1}^C \prod_{m=1}^{M_c} \left\{ p(\boldsymbol{\alpha}^{c,m}) p(\boldsymbol{\gamma}^{c,m}) p(\boldsymbol{\lambda}^{c,m}) \right. \\ &\quad \times \prod_{k=1}^K p(u_k^{c,m}) \prod_{n=1}^N [p(\mathbf{x}_n^{c,m}|\Theta, \Upsilon) p(\mathbf{b}_n^{c,m}|\Theta, \Upsilon) \\ &\quad \times p(\mathbf{w}_n^{c,m}|\boldsymbol{\alpha}^{c,m}) p(\mathbf{z}_n^{c,m}|\mathbf{u}_k^{c,m})] \left. \right\} \times \prod_{k=1}^K p(\mathbf{a}_k) p(\mathbf{d}_k). \end{aligned} \quad (\text{A.1})$$

According to (A.1), we can derive the conditional posterior distributions as follows.

1) The posterior distribution of the shared factor dictionary

$$\begin{aligned} \mathbf{A} &= [\mathbf{a}_1, \dots, \mathbf{a}_K]^T \\ p_{\mathbf{a}_k} &= N(\mathbf{a}_k | \boldsymbol{\zeta}_{\mathbf{a}_k}, \boldsymbol{\Sigma}_{\mathbf{a}_k}) \\ \boldsymbol{\zeta}_{\mathbf{a}_k} &= \boldsymbol{\Sigma}_{\mathbf{a}_k} \sum_{c=1}^C \sum_{m=1}^{M_c} \sum_{n=1}^N (z_{n,k}^{c,m})^T (w_{n,k}^{c,m})^T \boldsymbol{\gamma}^{c,m} \\ &\quad \times \left[\mathbf{x}_n^{c,m} - \boldsymbol{\mu}^{c,m} - \sum_{i=1, i \neq k}^K \mathbf{a}_i w_{n,i}^{c,m} z_{n,i}^{c,m} \right] \\ \boldsymbol{\Sigma}_{\mathbf{a}_k} &= \left[\sum_{c=1}^C \sum_{m=1}^{M_c} \sum_{n=1}^N (z_{n,k}^{c,m})^T (w_{n,k}^{c,m})^T \right. \\ &\quad \times \boldsymbol{\gamma}^{c,m} (z_{n,k}^{c,m}) (w_{n,k}^{c,m}) + P \left. \right]^{-1} \mathbf{I}_P \end{aligned} \quad (\text{A.2})$$

where $\mathbf{a}_k$ is the k th column of $\mathbf{A}$, $z_{n,k}^{c,m}$ is the k th atom of binary vector $\mathbf{z}_n^{c,m}$, $w_{n,k}^{c,m}$ is the k th atom of weight vector $\mathbf{w}_n^{c,m}$, and the mean vector $\boldsymbol{\mu}^{c,m}$ of the m th frame of the c th target is calculated by $\boldsymbol{\mu}^{c,m} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n^{c,m}$.

2) The posterior distribution of the shared label preserving dictionary $\mathbf{D} = [\mathbf{d}_1, \dots, \mathbf{d}_K]^T$

$$\begin{aligned} p_{\mathbf{d}_k} &= N(\mathbf{d}_k | \boldsymbol{\zeta}_{\mathbf{d}_k}, \boldsymbol{\Sigma}_{\mathbf{d}_k}) \\ \boldsymbol{\zeta}_{\mathbf{d}_k} &= \boldsymbol{\Sigma}_{\mathbf{d}_k} \sum_{c=1}^C \sum_{m=1}^{M_c} \sum_{n=1}^N (z_{n,k}^{c,m})^T (w_{n,k}^{c,m})^T \boldsymbol{\lambda}^{c,m} \\ &\quad \times \left[\boldsymbol{\varphi}_n^{c,m} - \sum_{i=1, i \neq k}^K \mathbf{d}_i w_{n,i}^{c,m} z_{n,i}^{c,m} \right] \\ \boldsymbol{\Sigma}_{\mathbf{d}_k} &= \left[\sum_{c=1}^C \sum_{m=1}^{M_c} \sum_{n=1}^N (z_{n,k}^{c,m})^T (w_{n,k}^{c,m})^T \boldsymbol{\lambda}^{c,m} \right. \\ &\quad \times (z_{n,k}^{c,m}) (w_{n,k}^{c,m}) + L \left. \right]^{-1} \mathbf{I}_L. \end{aligned} \quad (\text{A.3})$$

3) The posterior distribution of the weight vector $\mathbf{w}_n^{c,m}$

$$p_{\mathbf{w}_n^{c,m}} = N(\mathbf{w}_n^{c,m} | \boldsymbol{\zeta}_{\mathbf{w}_n^{c,m}}, \Sigma_{\mathbf{w}_n^{c,m}})$$

$$\boldsymbol{\zeta}_{\mathbf{w}_n^{c,m}} = \Sigma_{\mathbf{w}_n^{c,m}}$$

$$\left[\begin{aligned} & \gamma^{c,m} \text{diag}(\mathbf{z}_n^{c,m}) \mathbf{A}^T \left(\mathbf{x}_n^{c,m} - \boldsymbol{\mu}^{c,m} - \sum_{i=1, i \neq k}^K \mathbf{a}_i w_{n,i}^{c,m} z_{n,i}^{c,m} \right) \\ & + \lambda^{c,m} \text{diag}(\mathbf{z}_n^{c,m}) \mathbf{D}^T \left(\boldsymbol{\varphi}_n^{c,m} - \sum_{i=1, i \neq k}^K \mathbf{d}_i w_{n,i}^{c,m} z_{n,i}^{c,m} \right) \end{aligned} \right]^{-1}$$

$$\Sigma_{\mathbf{w}_n^{c,m}} = \left[\begin{aligned} & \gamma^{c,m} \text{diag}(\mathbf{z}_n^{c,m}) \mathbf{A}^T \mathbf{A} \text{diag}(\mathbf{z}_n^{c,m}) \\ & + \lambda^{c,m} \text{diag}(\mathbf{z}_n^{c,m}) \mathbf{D}^T \mathbf{D} \text{diag}(\mathbf{z}_n^{c,m}) + \alpha^{c,m} \mathbf{I}_K \end{aligned} \right]^{-1}. \quad (\text{A.4})$$

4) The posterior distribution of the binary vector $\mathbf{z}_n^{c,m}$

$$p_{z_{n,k}^{c,m}} = \text{Bern}(p(z_{n,k}^{c,m} = 1) / (p(z_{n,k}^{c,m} = 1) + p(z_{n,k}^{c,m} = 0)))$$

$$p(z_{n,k}^{c,m} = 0) \propto \exp[\ln(1 - u_k^{c,m})]$$

$$p(z_{n,k}^{c,m} = 1) \propto \exp\left[\ln(u_k^{c,m}) - 1/2(\tilde{\mathbf{x}}_n^{c,m})^T \gamma^{c,m} (\tilde{\mathbf{x}}_n^{c,m}) - 1/2(\tilde{\boldsymbol{\varphi}}_n^{c,m})^T \lambda^{c,m} (\tilde{\boldsymbol{\varphi}}_n^{c,m})\right] \quad (\text{A.5})$$

where $\tilde{\mathbf{x}}_n^{c,m} = \mathbf{x}_n^{c,m} - \boldsymbol{\mu}^{c,m} - \sum_{i=1, i \neq k}^K \mathbf{a}_i w_{n,i}^{c,m} z_{n,i}^{c,m} - \mathbf{a}_k w_{n,k}^{c,m}$, and $\tilde{\boldsymbol{\varphi}}_n^{c,m} = \boldsymbol{\varphi}_n^{c,m} - \sum_{i=1, i \neq k}^K \mathbf{d}_i w_{n,i}^{c,m} z_{n,i}^{c,m} - \mathbf{d}_k w_{n,k}^{c,m}$.

5) The posterior distribution of the label preserving constraint term $\boldsymbol{\varphi}_n^{c,m} = [\varphi_{n,1}^{c,m}, \dots, \varphi_{n,L}^{c,m}]^T$

$$p_{\varphi_{n,l}^{c,m}} = N(\boldsymbol{\zeta}_{\varphi_{n,l}^{c,m}}, \Sigma_{\varphi_{n,l}^{c,m}})$$

$$\boldsymbol{\zeta}_{\varphi_{n,l}^{c,m}} = (b_{n,l}^{c,m} - 1/2 + \boldsymbol{\zeta}_{\varphi_{n,l}^{c,m}} \Sigma_{\varphi_{n,l}^{c,m}}) / (\delta_{n,l}^{c,m} + \Sigma_{\varphi_{n,l}^{c,m}})$$

$$\Sigma_{\varphi_{n,l}^{c,m}} = 1 / (\delta_{n,l}^{c,m} + \Sigma_{\varphi_{n,l}^{c,m}}) \quad (\text{A.6})$$

where $\varphi_{n,l}^{c,m}$ is the l th atom of $\boldsymbol{\varphi}_n^{c,m}$ and $\delta_{n,l}^{c,m}$ can be sampled from the posterior distribution $p_{\delta_{n,l}^{c,m}} = \text{PG}(1, \varphi_{n,l}^{c,m})$.

6) The posterior distribution of the weight vector parameter $\alpha^{c,m}$

$$p_{\alpha^{c,m}} = \text{Ga}(\tilde{a}_0, \tilde{b}_0)$$

$$\tilde{a}_0 = a_0 + 1/2N K$$

$$\tilde{b}_0 = b_0 + 1/2 \sum_{n=1}^N (\mathbf{w}_n^{c,m})^T \mathbf{w}_n^{c,m}. \quad (\text{A.7})$$

7) The posterior distribution of the binary vector parameter $u_k^{c,m}$

$$p_{u_k^{c,m}} = \text{Beta}(\tilde{c}_0, \tilde{d}_0)$$

$$\tilde{c}_0 = c_0 + \sum_{n=1}^N z_{n,k}^{c,m}$$

$$\tilde{d}_0 = d_0 + N - \sum_{n=1}^N z_{n,k}^{c,m}. \quad (\text{A.8})$$

8) The posterior distribution of the noise parameter $\gamma^{c,m}$

$$p_{\gamma^{c,m}} = \text{Ga}(\tilde{e}_0, \tilde{f}_0)$$

$$\tilde{e}_0 = e_0 + 1/2PN$$

$$\tilde{f}_0 = f_0 + \sum_{n=1}^N 1/2 [\mathbf{x}_n^{c,m} - \boldsymbol{\mu}^{c,m} - \mathbf{A}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m})]^T \times [\mathbf{x}_n^{c,m} - \boldsymbol{\mu}^{c,m} - \mathbf{A}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m})]. \quad (\text{A.9})$$

9) The posterior distribution of the noise parameter $\lambda^{c,m}$

$$p_{\lambda^{c,m}} = \text{Ga}(\tilde{g}_0, \tilde{h}_0)$$

$$\tilde{g}_0 = g_0 + 1/2LN$$

$$\tilde{h}_0 = h_0 + \sum_{n=1}^N 1/2 [\boldsymbol{\varphi}_n^{c,m} - \mathbf{D}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m})]^T \times [\boldsymbol{\varphi}_n^{c,m} - \mathbf{D}(\mathbf{w}_n^{c,m} \odot \mathbf{z}_n^{c,m})]. \quad (\text{A.10})$$

With the above conditional posterior distributions, a Markov chain is constructed. At each iteration, the samples are drawn from the above conditional posterior distributions.

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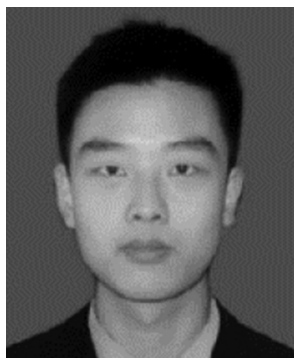
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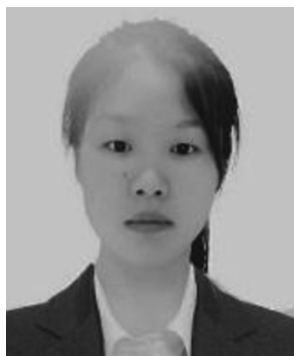
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